CEBU INSTITUTE OF TECHNOLOGY UNIVERSITY

COLLEGE OF COMPUTER STUDIES

Software Requirements Specifications

for

PRISM

Change History

Version	Date	Author	Description
1.0	5/6/2026	Pangilinan, Vein Carmell G.	Initial Draft of SRS document
1.2	5/6/2026	Uy, Emman Jay C.	Initial diagrams for UC-1.3 and 2.1
1.1	5/7/2026	Binagatan, Alexander Jr.	Added use case diagrams and activity diagrams for UC 1.1 and 1.2
1.3	5/7/2026	Pangilinan, Vein Carmell G.	Added use case diagrams and activity diagrams for UC 3.1 and 3.2
1.4	5/8/2026	Atamosa, Charry Mae	Added use case diagrams and activity diagrams for UC 3.3 and 4.1
1.5	5/8/2026	Uy, Emman Jay C.	Final diagrams for UC-1.3 and 2.1
1.6	5/10/2026	Pangilinan, Vein Carmell G.	Added wireframes for UC 3.1 and 3.2
1.7	5/10/2026	Atamosa, Charry Mae	Added wireframes for UC 3.3 and 4.1
1.8	5/10/2026	Binagatan, Alexander Jr.	Added wireframes for UC 1.1 and 1.2
1.9	5/11/2026	Uy, Emman Jay C.	Added wireframes for UC 1.3 and 2.1
1.10	5/11/2026	Beato, Angel Anel	Added non-functional requirements, wireframes, use case diagram, activity diagram for UC 4.2

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1. Introduction

1.1. Purpose

This Software Requirements Specifications (SRS) document provides a complete and traceable description of PRISM — the official one-word project title for the Parenteral Return-demonstration Injection Skills Monitor. PRISM is an Android mobile application developed using the Flutter framework that applies MediaPipe-based computer vision and the Llama 3.3 large language model (via OpenRouter API) to objectively evaluate three safety-critical procedural components of nursing parenteral injection return demonstrations at the Cebu Institute of Technology – University (CIT-U).

1.2. Scope

PRISM is scoped as a mobile Android application deployed within CIT-U's nursing skills laboratory. It addresses the documented unreliability of manual visual assessment during parenteral injection return demonstrations, where human observers misjudge insertion angles by 10–15 degrees due to parallax effects (Walsh et al., 2021).

- Real-time computer vision detection of needle insertion angle, aspiration technique, and needle withdrawal angle across four injection types (IM: 90°, SubQ: 45°, IV: 15°, ID: 10°)
- AI-generated, clinically worded performance feedback via Llama 3.3 (OpenRouter API)
- Role-based access for nursing students and clinical instructors via Firebase Authentication
- Session record storage and longitudinal review via Firebase Firestore
- Instructor feedback review and controlled release to students

The system supplements, and does not replace, instructor judgment. Tactile components (e.g., force application, aseptic technique) and non-measurable procedural steps are outside scope.

1.3. Definitions, Acronyms and Abbreviations

Term / Acronym	Definition
PRISM	The official 1-word project title and product name. Stands for ParenteralReturn-demonstration Injection Skills Monitor.
IM	Intramuscular — parenteral injection administered at 90°.
SubQ / SC	Subcutaneous — parenteral injection administered at 45°.
IV	Intravenous — parenteral injection administered at 15°.
ID	Intradermal — parenteral injection administered at 10°.
MAE	Mean Absolute Error — the primary metric for angle detection accuracy (target: ≤ 3°).

MediaPipe	Google’s open-source ML framework for real-time on-device hand landmark detection.
Landmark 0	Wrist landmark in the MediaPipe Hand Landmarker 21-point model, used as the base point for angle computation.
Landmark 8	Index fingertip landmark in the MediaPipe model, used as the distal point for angle vector computation.
Landmark 4	Thumb tip landmark in the MediaPipe model, used for aspiration plunger withdrawal tracking.
Llama 3.3	Meta’s Llama 3.3 70B Instruct large language model, accessed via the OpenRouter API for clinical feedback generation.
OpenRouter	API gateway used to access Llama 3.3 for AI feedback generation via REST/HTTPS.
Firebase Auth	Firebase Authentication — cloud-based user identity and role management service.
Firestore	Firebase Firestore — NoSQL cloud database for persistent session record storage.
Flutter	Google’s open-source UI toolkit used to develop the Android application.
Return Demo	Return demonstration — a nursing competency assessment where students perform a procedure for instructor evaluation.

1.4. References

- Walsh, C., Harder, N., & Hollingsworth, H. (2021). Rater agreement and the assessment of injection angle in nursing simulation. Nurse Education Today, 98, 104723.
- Chen, L.-C., et al. (2023). Clinical instructors’ perspectives on the assessment of clinical knowledge. Healthcare, 11, 1851.
- Tantacharoenrat, C., & Precharattana, M. (2024). Survey of learning experience of pediatric injection. Journal of Education and Health Promotion, 13, 226.
- Maino, C., et al. (2024). Hand tracking for clinical applications: Validation of MediaPipe Hand frameworks. Biomedical Signal Processing and Control.
- Saccenti, L., et al. (2025). Integrated needle guide on smartphone for percutaneous interventions. CardioVascular and Interventional Radiology.
- George, T. P., DeCristofaro, C., & Ford Murphy, P. (2020). Self-efficacy and concerns of nursing students regarding clinical experiences. Nurse Education Today, 90, 104401.

Overall Description

1.5. Product perspective

PRISM operates as a standalone Android mobile application integrating three external services: (1) MediaPipe Hand Landmarker for on-device, marker-free computer vision; (2) Llama 3.3 70B Instruct via the OpenRouter API for AI-generated clinical feedback; and (3) Firebase (Authentication + Firestore) for user identity management and cloud-based session record storage.

PRISM addresses a documented gap in nursing education at CIT-U: the absence of any objective, technology-based measurement tool aligned to institutional competency standards. Unlike sensor-based alternatives (e.g., IMU-embedded syringes), PRISM relies exclusively on consumer smartphone hardware, removing cost and infrastructure barriers. The system digitizes the CIT-U five-point rubric into a data-driven assessment pipeline, providing immediate feedback unavailable in current manual practice. Clinical instructors retain final review and release authority over all AI-generated outputs before students can access them.

1.6. User characteristics

Nursing Students

- Undergraduate nursing students at CIT-U completing parenteral injection return demonstrations as part of their curriculum.
- Expected to have basic smartphone literacy; no prior experience with computer vision or AI-assisted assessment required.
- Goals: receive immediate structured feedback, track performance over time, and use feedback to improve procedural accuracy before clinical placement.

Clinical Instructors

- Licensed nursing faculty at CIT-U responsible for evaluating student return demonstrations against CIT-U rubric checklists.
- Experienced with manual assessment workflows; may have limited prior exposure to AI-assisted tools.
- Goals: reduce scoring variability and workload, access centralized student performance records, review and validate AI feedback before release, and retain professional judgment authority.

2.4. Constraints

- Platform: Android 8.0 (API Level 26) or higher only. iOS is out of scope for this project cycle.
- Hardware: Minimum 8 MP rear-facing camera with autofocus; minimum 3 GB RAM for real-time MediaPipe inference; minimum 1080×1920 display resolution.
- Environmental: Adequate lighting, stable camera placement, and uncluttered background required for reliable landmark detection. Occlusion of hand or syringe degrades detection.
- Network: Firebase Authentication, Firestore sync, and OpenRouter API calls require stable 4G LTE or Wi-Fi. MediaPipe runs on-device and does not require connectivity.
- Scope: Only three measurable procedural variables are evaluated (insertion angle, aspiration, withdrawal angle). Tactile components and full injection procedure assessment are out of scope.
- Data Privacy: No session video footage is stored. Only computed metrics and AI-generated text are persisted in Firestore.
- Assessment Authority: AI-generated feedback is advisory. Instructors must review and release all feedback before students can access it.

2.5. Assumptions and dependencies

- CIT-U will provide skills laboratory access for UAT with a minimum of five nursing students and two clinical instructors.
- CIT-U clinical instructors will supply expert-annotated video baselines for technical accuracy validation.
- Test devices run Android 8.0+ with rear-facing cameras of at least 8 MP.
- The OpenRouter API will maintain Llama 3.3 70B Instruct availability throughout the development and testing period.
- Firebase services remain available and the team maintains valid API credentials for the project duration.
- The student's dominant hand and syringe are visible to the camera with sufficient clarity for MediaPipe to reliably track Landmarks 0, 4, and 8.
- Users will position the smartphone camera perpendicular to the injection plane to minimize parallax error.
- Performance targets (MAE $\leq 3°$, SUS ≥ 80) are contingent on the availability of representative annotated video datasets.

2. Specific Requirements

2.1. External interface requirements

3.1.1. Hardware interfaces

- Smartphone Rear Camera: Primary sensor for real-time video capture, accessed via the Flutter camera plugin. Minimum: 8 MP with autofocus. Camera feed is processed entirely on-device by MediaPipe.
- Device Display: Renders camera preview, landmark overlays, angle annotations, and application UI. Minimum: Full HD (1080×1920 px).
- Device Processor: MediaPipe Hand Landmarker performs on-device ML inference. Minimum: ARM64 CPU, 3 GB RAM.
- Touchscreen Input: All user interactions (type selection, session control, navigation) via capacitive touchscreen.

3.1.2. Software interfaces.

- Flutter (v3.x): Cross-platform mobile development framework targeting Android.
- MediaPipe Hand Landmarker: On-device hand tracking providing 21 landmarks per frame. Key landmarks: 0 (wrist), 8 (index fingertip), 4 (thumb tip). Integrated via the mediapipe Flutter plugin.
- OpenRouter API / Llama 3.3 70B Instruct: REST API endpoint for AI feedback generation. HTTPS POST with JSON payload. Expected response: ≈2 seconds.
- Firebase Authentication: Manages registration, login, session tokens, and role claims (Student / Instructor).
- Firebase Firestore: NoSQL cloud database storing session records (metrics, AI feedback text, scores, timestamps, user IDs). No video data stored.
- Android SDK (API Level 26+): Target deployment platform via the Flutter Android build pipeline.

3.1.3. Communications interfaces

- HTTPS / REST: All external communications (OpenRouter, Firebase Auth, Firestore) use secure HTTPS with JSON exchange.
- Firebase SDK Listeners: Firestore snapshot listeners enable near-real-time dashboard updates when new session records are created.
- Internet Connectivity: Minimum 4G LTE or stable Wi-Fi required for cloud-dependent features. On-device MediaPipe inference requires no network access.

2.2. Functional requirements

PRISM is structured into four modules, each corresponding to one general objective from the approved proposal. The specific objectives within each general objective are translated into transactions and use cases under their respective module.

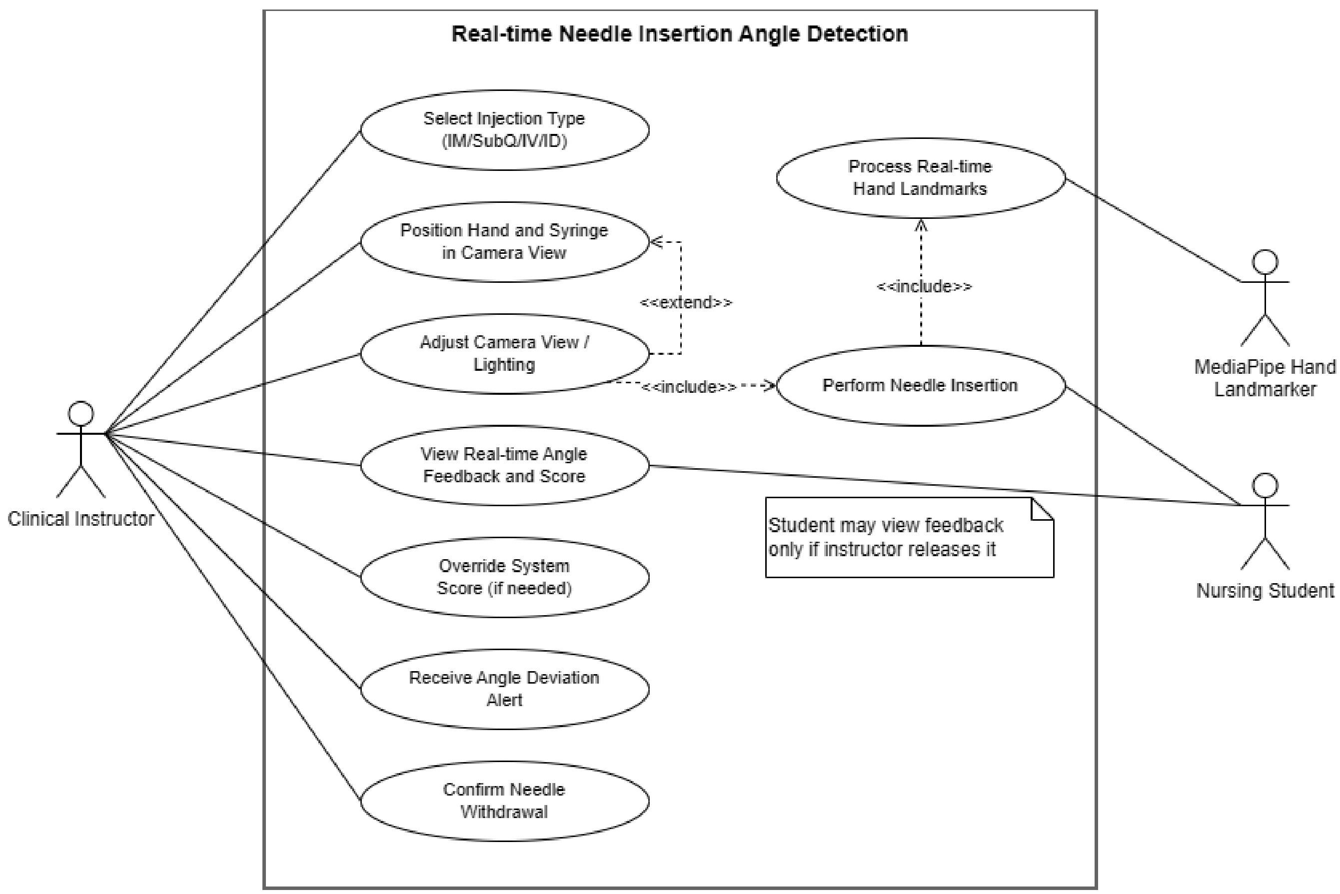
Module 1

Corresponds to General Objective 1: To design and develop PRISM as a mobile Android application using Flutter that utilizes MediaPipe-based computer vision to detect needle insertion angle, aspiration technique, and needle withdrawal angle across four parenteral injection types, targeting MAE within three degrees of expert-annotated baselines.

Specific Objectives covered: 1.1 (insertion angle detection), 1.3 (aspiration technique detection), 1.4 (withdrawal angle detection).

UC-1.1:

Use Case Diagram

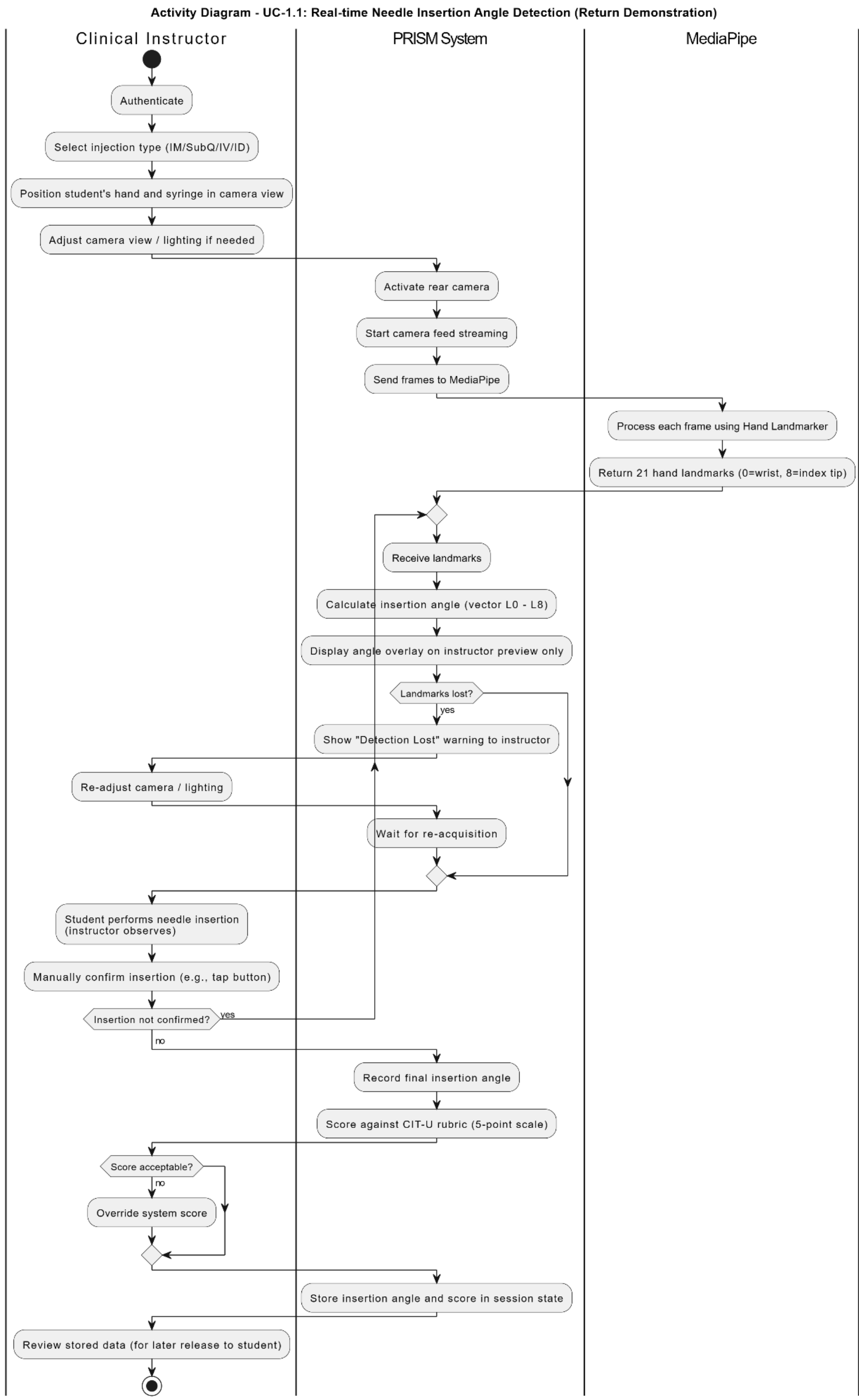


Use Case Description

Use Case ID	*UC-1.1*
Use Case Name	*Real-time Needle Insertion Angle Detection*
Module	*Module 1: Detection Engine*

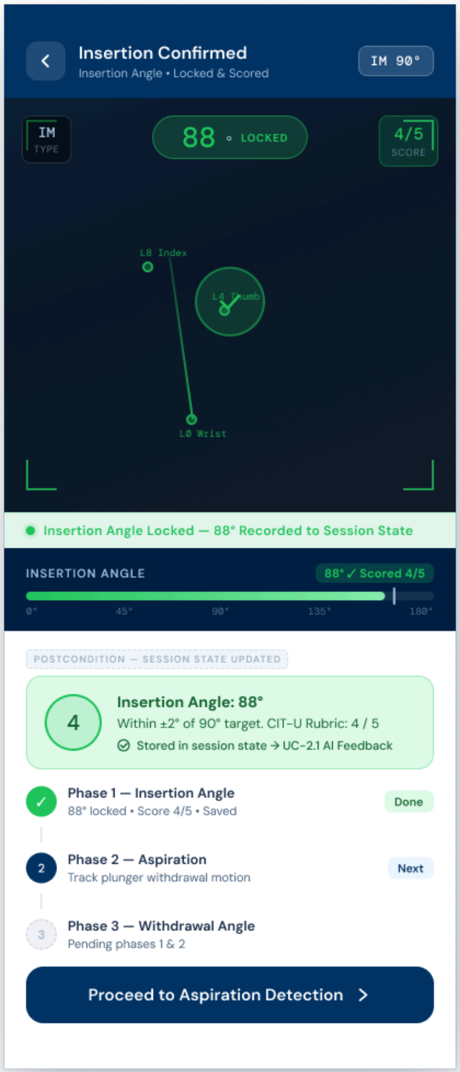
Actors	*Clinical Instructor (primary — configures and monitors); Nursing Student (primary — performs procedure); System (MediaPipe Hand Landmarker)*
Preconditions	*1. Clinical Instructor is authenticated.* *2. An injection type has been selected and configured with target rubric thresholds either independently by the Student (Practice Mode via UC-3.2) or assigned via an Instructor session profile (Exam Mode)* *3. Instructor positions the student's hand and syringe in camera view, adjusts lighting as needed.* *4. The rear camera is active and the hand/syringe are within frame.* *5. Student is ready to perform the injection.*
Main Flow	*1. The user initializes the session transaction sequence on the PRISM application client.* *2. The system streams live camera frames to MediaPipe Hand Landmarker.* *3. MediaPipe computes 21 hand landmarks per frame in real time.* *4. The system calculates the insertion angle using vector-based geometric analysis: computing the absolute angle of the vector formed between Landmark 0 (wrist) and Landmark 8 (index fingertip) relative to the fixed horizontal or vertical grid axis of the smartphone camera sensor.* *5. The computed angle is displayed as an overlay on the instructor's and (optionally) student's preview, updating in real time.* *6. Student performs needle insertion.* *7. Upon confirmed insertion, the system locks and records the insertion angle value.* *8. The recorded angle is scored against the CIT-U five-point rubric for the selected injection type (IM: 90°, SubQ: 45°, IV: 15°, ID: 10°).* *9. Instructor can view the real-time angle feedback and score, and may override the system score if needed.*
Alternative Flow	*A1 – Landmarks lost (occlusion/poor lighting): System displays a 'Detection Lost' warning overlay and pauses angle recording until landmarks are re-acquired. Instructor may adjust camera or lighting.* *A2 – Angle deviation alert: If the student's insertion angle deviates beyond acceptable tolerance, the system sends a real-time alert to both instructor and student.* *A3 – Instructor override: After the session, instructor may manually adjust the system-generated score using the feedback review module (UC-4.2).*
Postconditions	*Insertion angle value and rubric score are stored in session state for use by UC-2.1 (AI Feedback Generation). Student may view feedback only after instructor releases it.*
Related SOs	*Specific Objective 1.1*
Priority	*High*

Activity Diagram



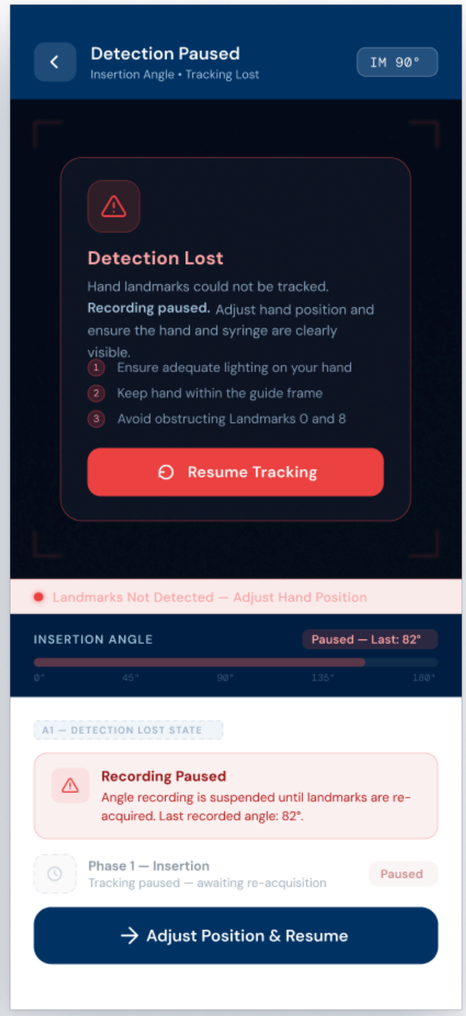
Wireframe

Insertion Confirmed Screen



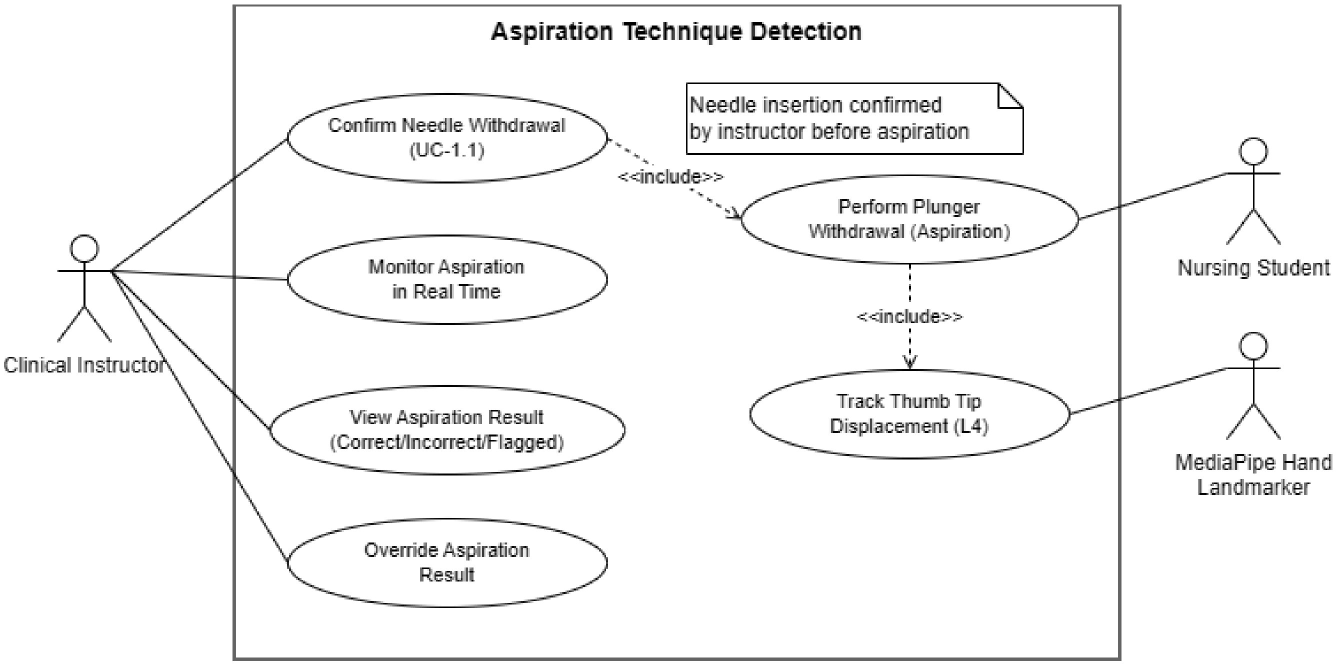
Real Time Detection Screen

Detection Lost Screen



UC-1.2: Aspiration Technique Detection

Use Case Diagram

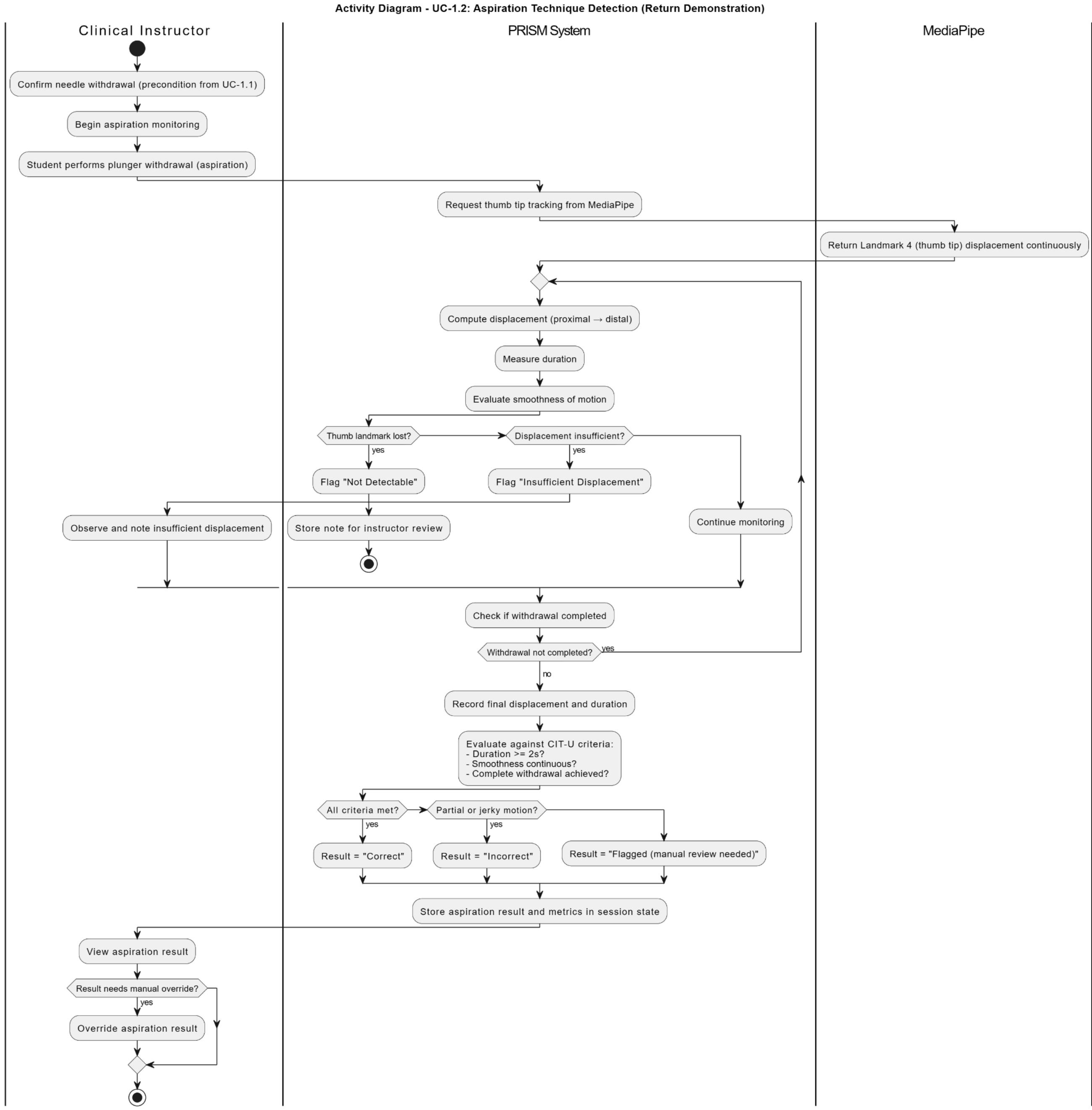


Use Case Description

Use Case ID	*UC-1.2*
Use Case Name	*Aspiration Technique Detection*
Module	*Module 1: Detection Engine*
Actors	*Clinical Instructor (primary – monitors, reviews, overrides), Nursing Student (primary – performs plunger withdrawal), MediaPipe Hand Landmarker (external system – provides thumb tip landmarks)*
Preconditions	*1. Needle insertion has been confirmed by the instructor (UC-1.1).* *2. The camera module is active and the student's thumb tip (Landmark 4) is visible (unless occluded, triggering alternative flow A1).*

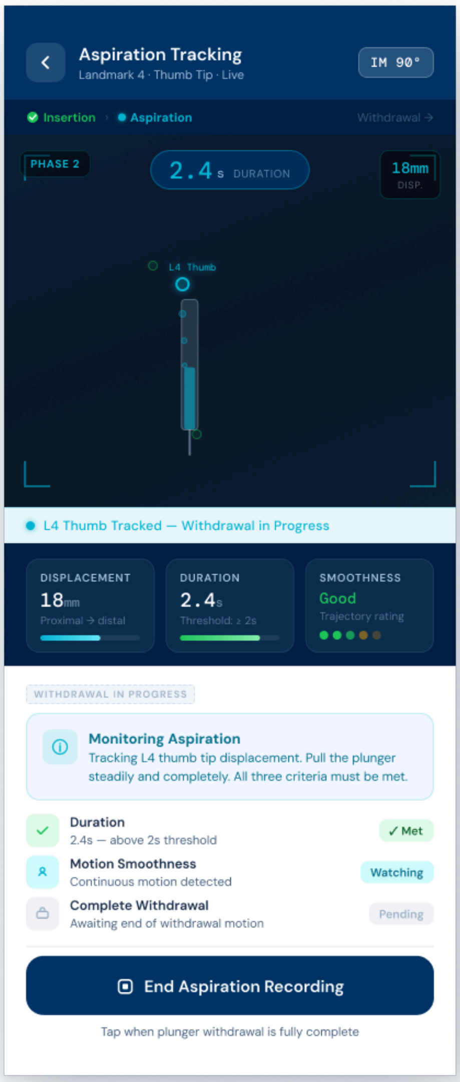
Main Flow	*1. The instructor monitors the aspiration phase in real time using the PRISM dashboard. 2. The student performs plunger withdrawal (aspiration). 3. The system tracks the coordinate displacement of the thumb tip (Landmark 4) via MediaPipe. **To normalize for camera-to-subject distance variations, the system computes a pixel-to-millimeter scaling ratio dynamically using a known anatomical reference constant, such as the pixel distance between the user's static metacarpophalangeal knuckles (Landmarks 5 and 17). 4. The system evaluates three criteria: a. Displacement (proximal → distal distance converted via scaling ratio, targeting e.g., ≥10mm) b. Duration (elapsed time, e.g., ≥2 seconds) c. Motion smoothness (trajectory steadiness) 5. The system determines the binary classification: Correct, Incorrect, or Flagged (if criteria partially met). 6. The instructor views the aspiration result (Correct/Incorrect/Flagged). 7. If needed, the instructor may override the system result (e.g., manual correction). 8. The final result, motion control rating, duration, and displacement are stored in session state.*
Alternative Flow	*A1 – Thumb landmark occluded (not detectable): System flags aspiration as “Not Detectable”, automatically appends a note for instructor review, and stores the flag in the session record A2 – Insufficient displacement (e.g., <10mm): System displays a warning (“Pull plunger further”), continues tracking, and does not finalise the result until sufficient displacement is detected or instructor intervenes*
Postconditions	*Aspiration result (including flags and instructor overrides) and all supporting metrics are stored in session state for later AI feedback generation (UC-2.1). The session record is marked as pending instructor review if any flag was raised.*
Related SOs	*Specific Objective 1.3*
Priority	*High*

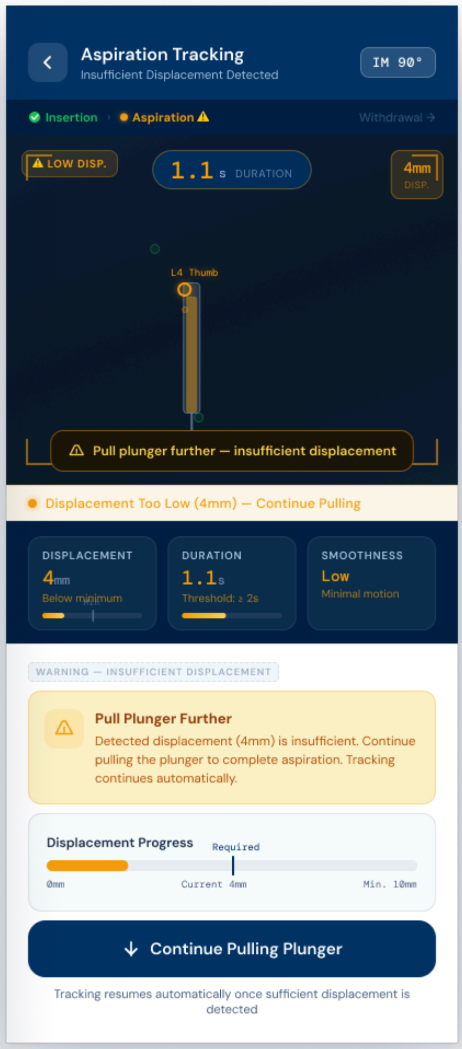
Activity Diagram

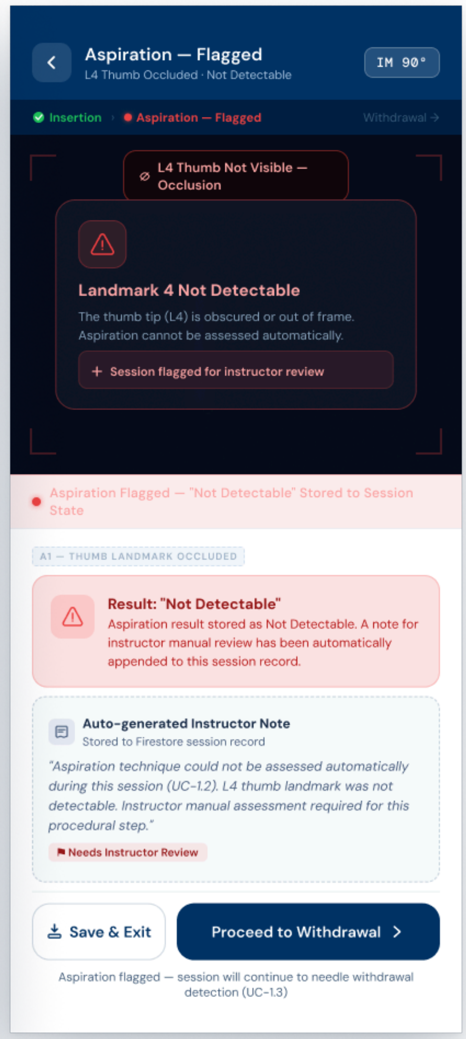


Wireframe

Aspiration Detection Active Screen Aspiration Insufficient Displacement Screen Thumb Not Detectable Screen

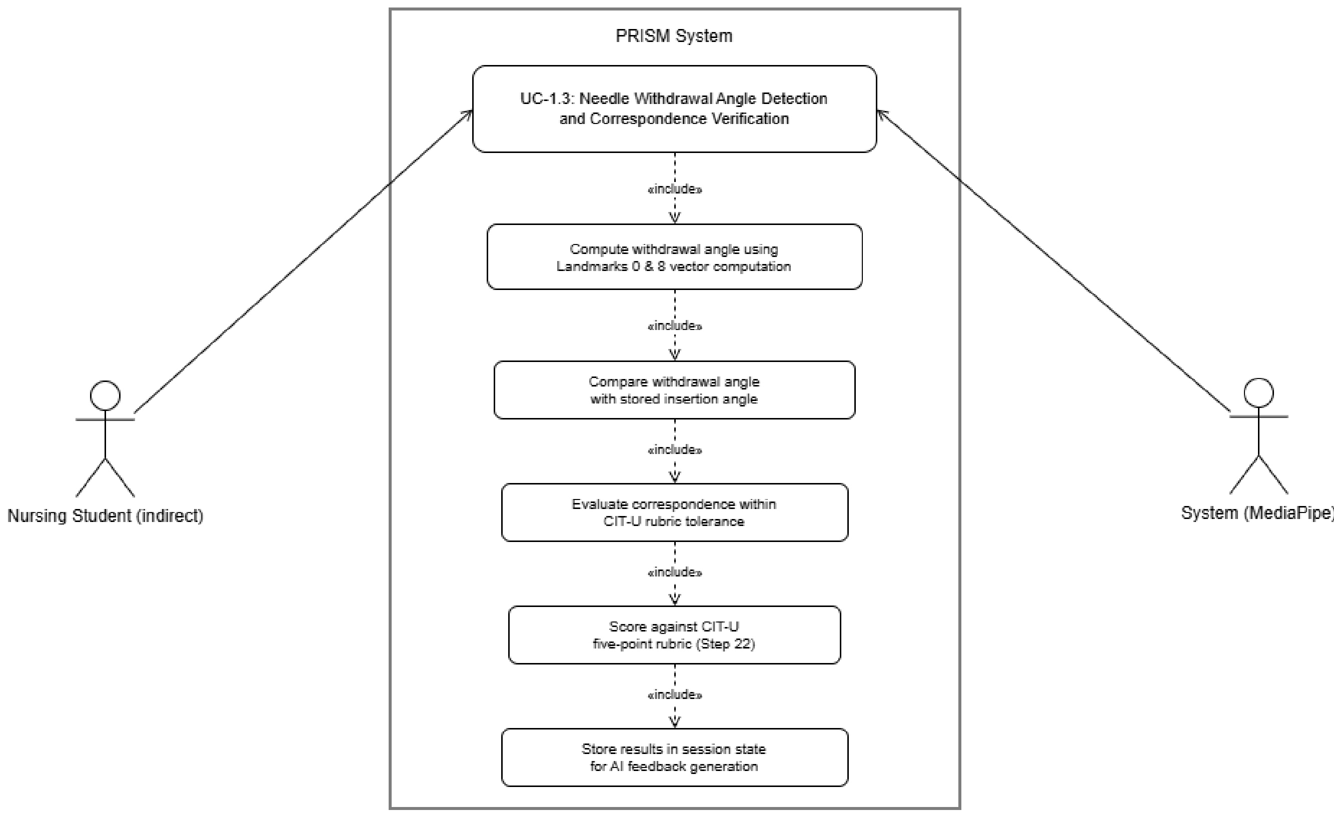






UC-1.3: Needle Withdrawal Angle Detection and Correspondence Verification

Use Case Diagram

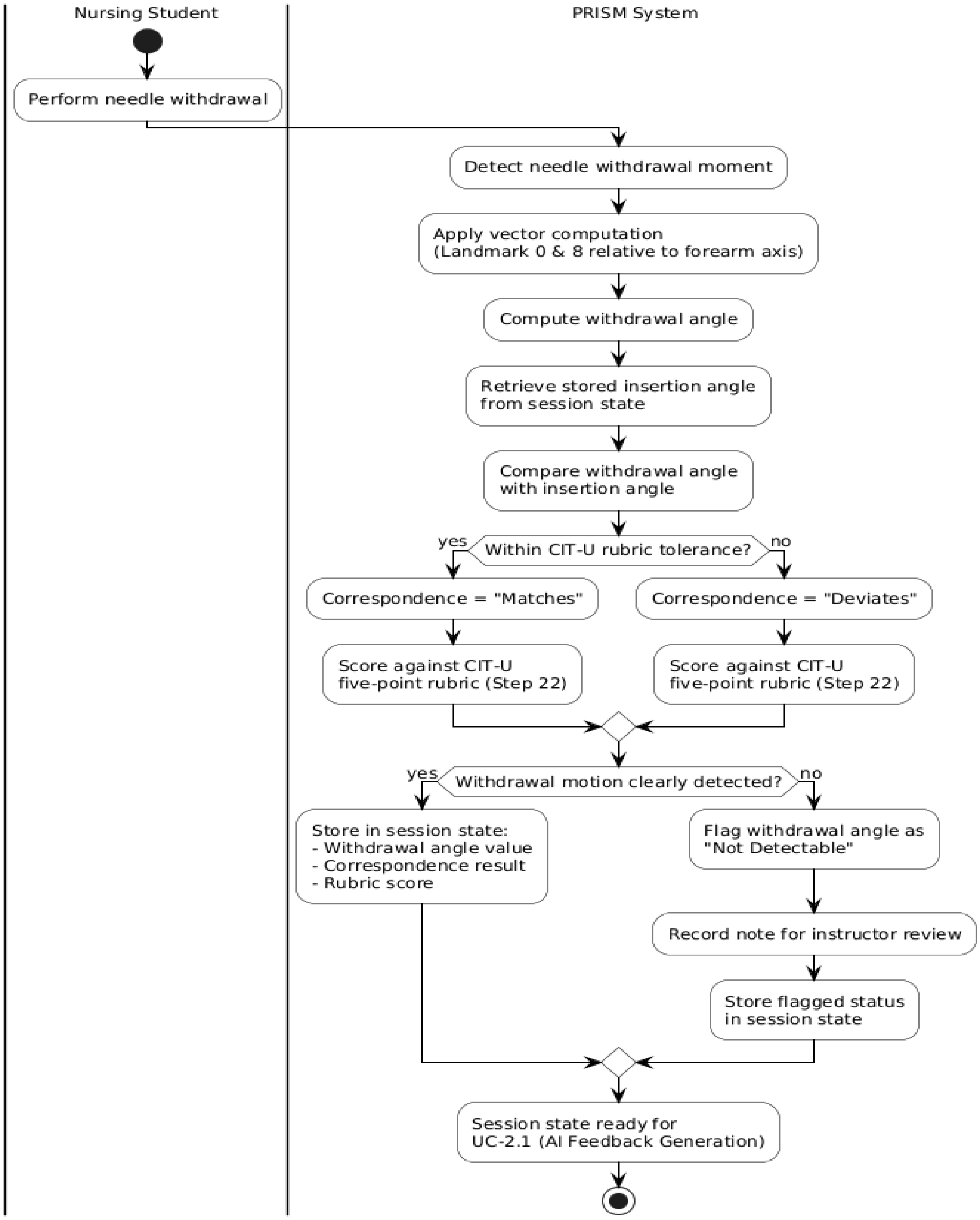


Use Case Description

Use Case ID	*UC-1.3*
Use Case Name	*Needle Withdrawal Angle Detection and Correspondence Verification*
Module	*Module 1: Detection Engine*
Actors	*Nursing Student (indirect); System (MediaPipe)*
Preconditions	*1. UC-1.1 has completed and insertion angle has been recorded in session state.* *2. UC-1.2 aspiration detection has completed.* *3. Camera module remains active.*

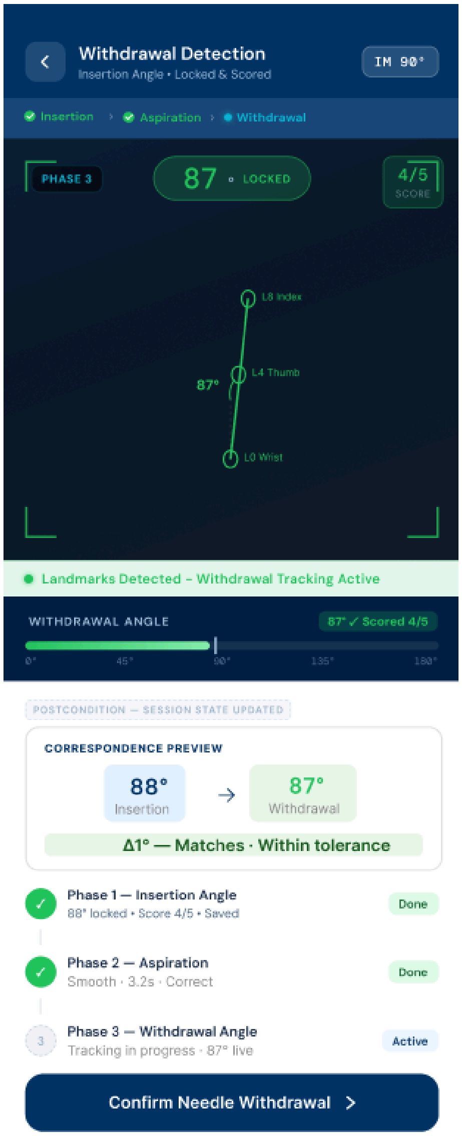
Main Flow	*1. At the moment of needle withdrawal, the system applies the same vector computation framework verified in UC-1.1, computing the absolute angle of the vector formed between Landmark 0 (wrist) and Landmark 8 (index fingertip) relative to the fixed horizontal or vertical grid axis of the smartphone camera sensor to calculate the withdrawal angle.* *2. The withdrawal angle is recorded and compared against the previously stored insertion angle.* *3. The system evaluates correspondence: whether the withdrawal angle matches the insertion angle within the CIT-U rubric tolerance.* *4. A correspondence result (Matches / Deviates) is computed and scored against the CIT-U five-point rubric (Rubric Step 22).*
Alternative Flow	*A1 — Withdrawal motion not clearly detected: System flags withdrawal angle as 'Not Detectable' and records a note for instructor review.*
Postconditions	*Withdrawal angle value, correspondence result, and rubric score are stored in session state for AI feedback generation.*
Related SOs	*Specific Objective 1.4*
Priority	*High*

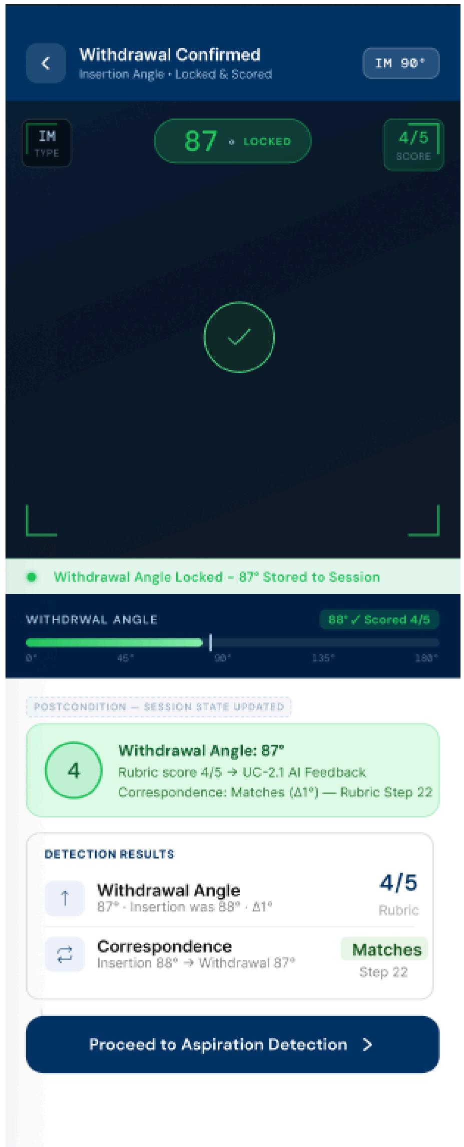
Activity Diagram

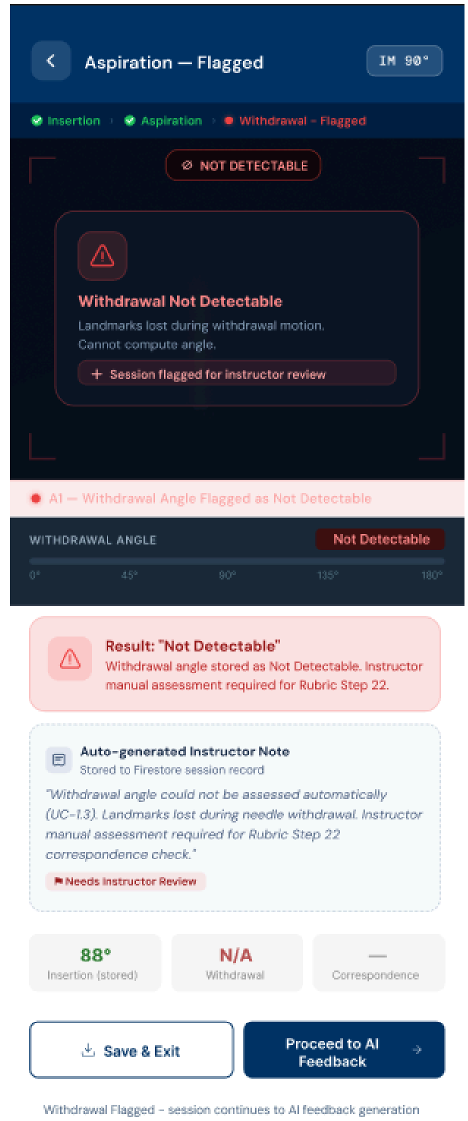


Wireframe

Needle Withdrawal Detection	Withdrawal Confirmed & Scored	Withdrawal Not Detectable







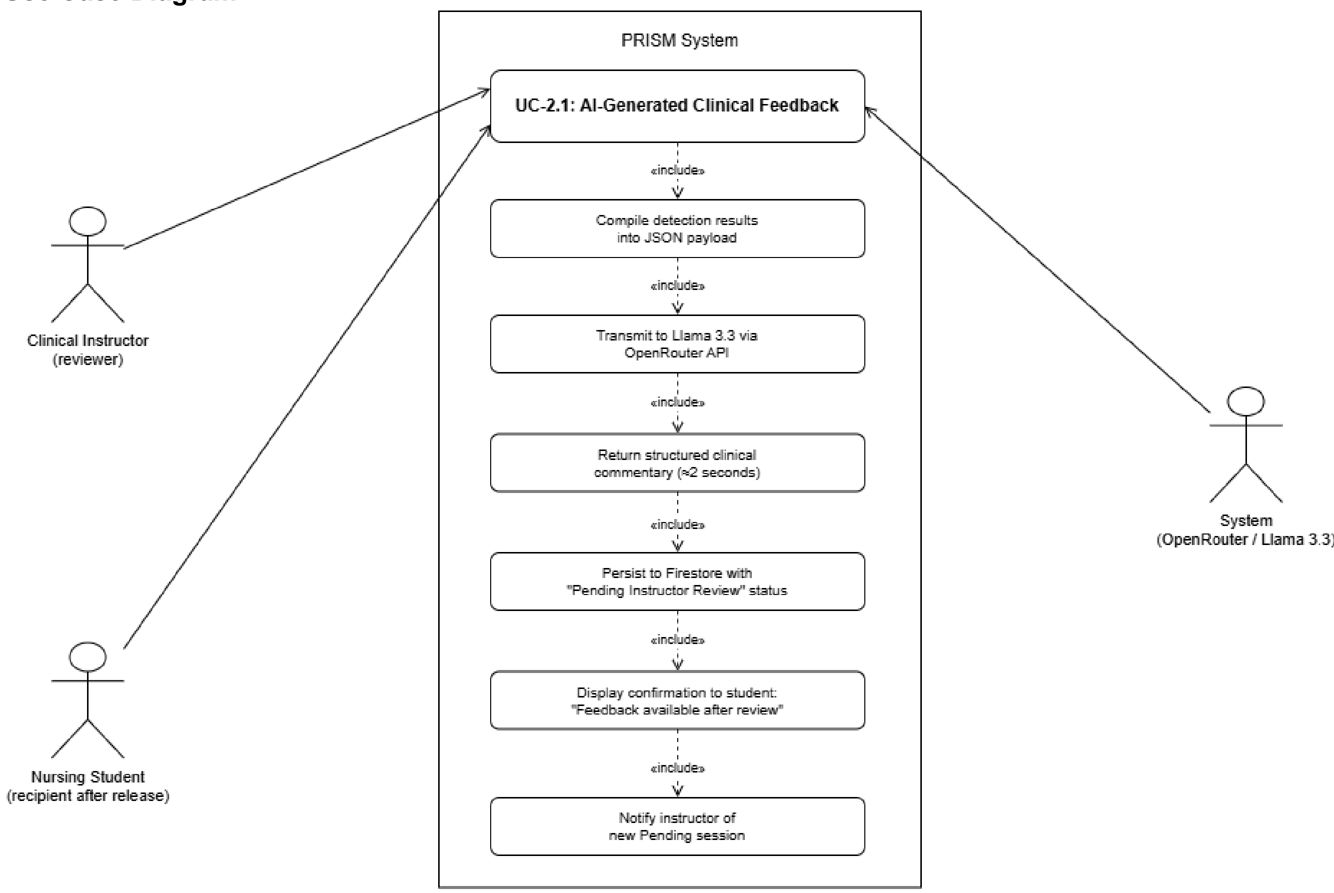
Module 2

Corresponds to General Objective 2: To integrate Llama 3.3 via the OpenRouter API within PRISM to process compiled detection results and generate structured, clinically worded performance feedback made available upon session completion.

Specific Objective covered: 2.1 (AI feedback generation and display).

UC-2.1: AI-Generated Clinical Feedback

Use Case Diagram

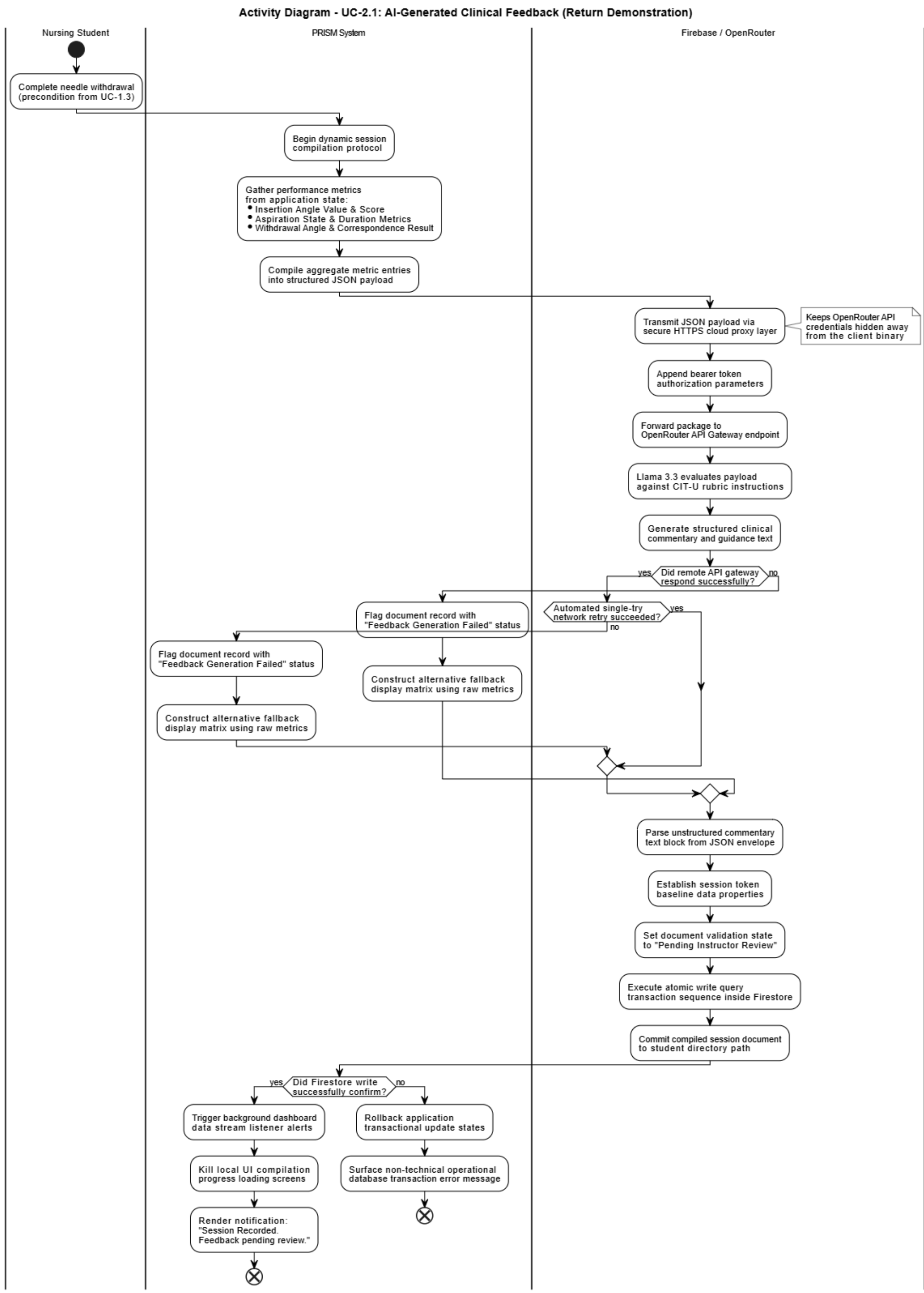


Use Case Description

Use Case ID	*UC-2.1*
Use Case Name	*AI-Generated Clinical Feedback*
Module	*Module 2: AI Feedback Engine*
Actors	*System (OpenRouter / Llama 3.3); Clinical Instructor (reviewer); Nursing Student (recipient after release)*

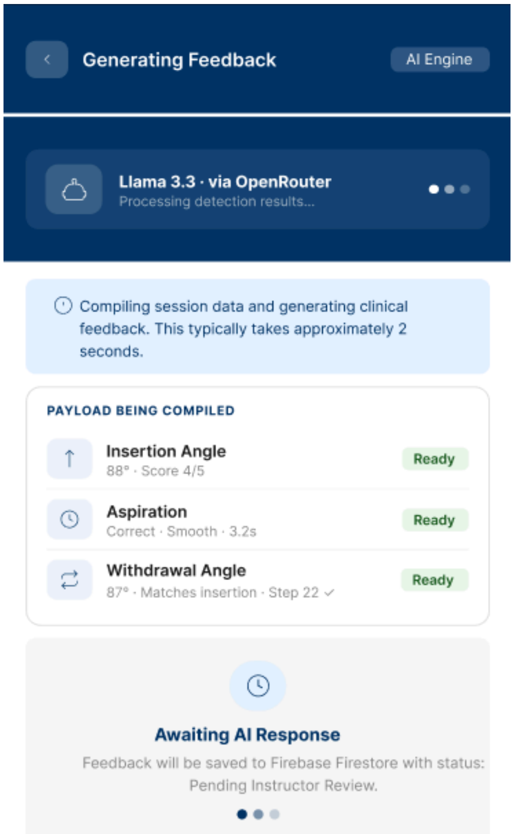
Preconditions	*1. UC-1.1, UC-1.2, and UC-1.3 have all completed.* *2. Session state contains: insertion angle value and score, aspiration result and metrics, withdrawal angle value and correspondence result.*
Main Flow	*1. Upon session completion, the system compiles all detection results from session state into a structured JSON payload.* *2. The system transmits the payload to Llama 3.3 70B Instruct via the OpenRouter API with a clinical feedback generation prompt aligned to CIT-U rubric standards.* *3. Llama 3.3 processes the data and returns structured, clinically worded performance commentary and improvement guidance within approximately 2 seconds.* *4. The AI-generated feedback is persisted to Firebase Firestore with status: Pending Instructor Review.* *5. The student's session results view displays: 'Your session has been recorded. Feedback will be available after instructor review.'* *6. The instructor is notified of the new Pending session (visible on the Dashboard, UC-4.1).*
Alternative Flow	*A1 — OpenRouter API timeout: System retries once. If unsuccessful, raw detection results are stored with a 'Feedback Generation Failed' flag. Instructor is notified via Dashboard.* *A2 — API returns malformed response: System stores raw metrics and flags the session for manual instructor feedback entry.*
Postconditions	*AI-generated feedback stored in Firestore with Pending status. Session record is complete and awaiting instructor review (UC-4.2).*
Related SOs	*Specific Objective 2.1*
Priority	*High*

Activity Diagram

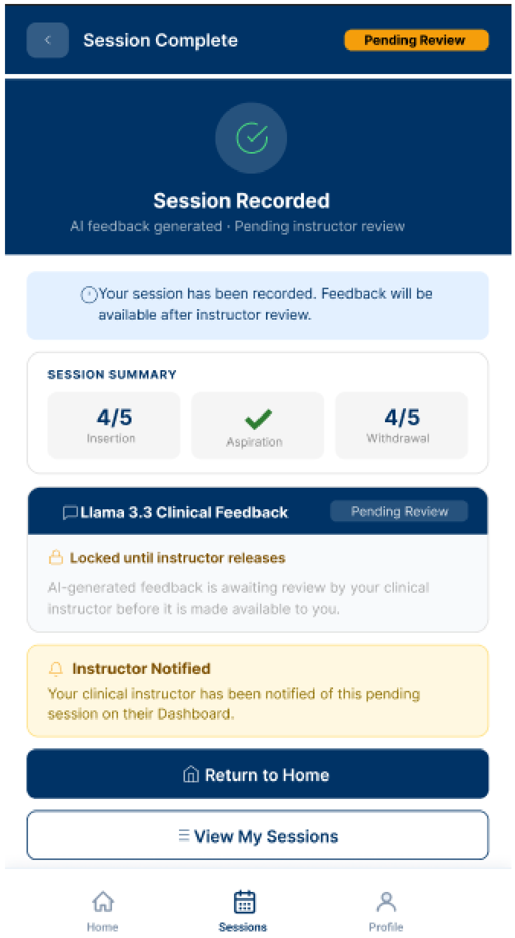


Wireframe

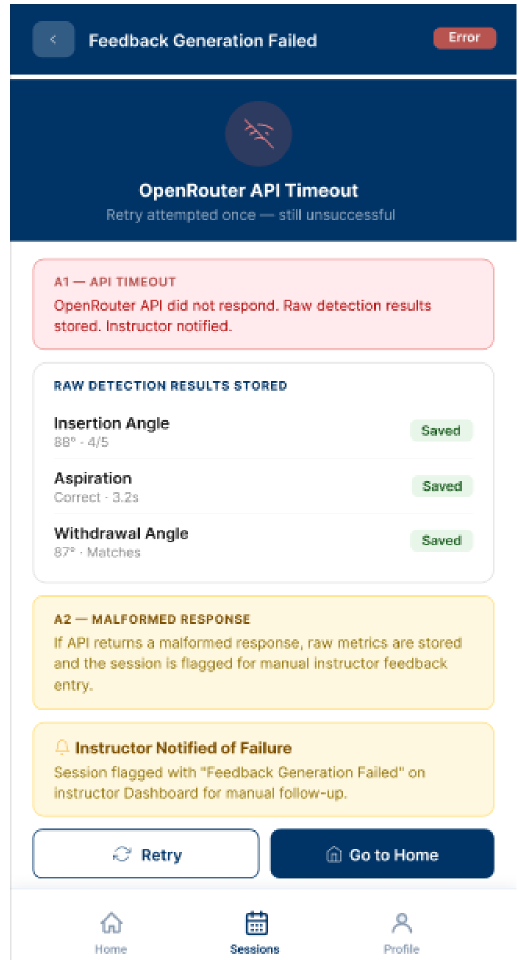
AI Feedback Generating



Session Recorded, Pending Review



API Error States



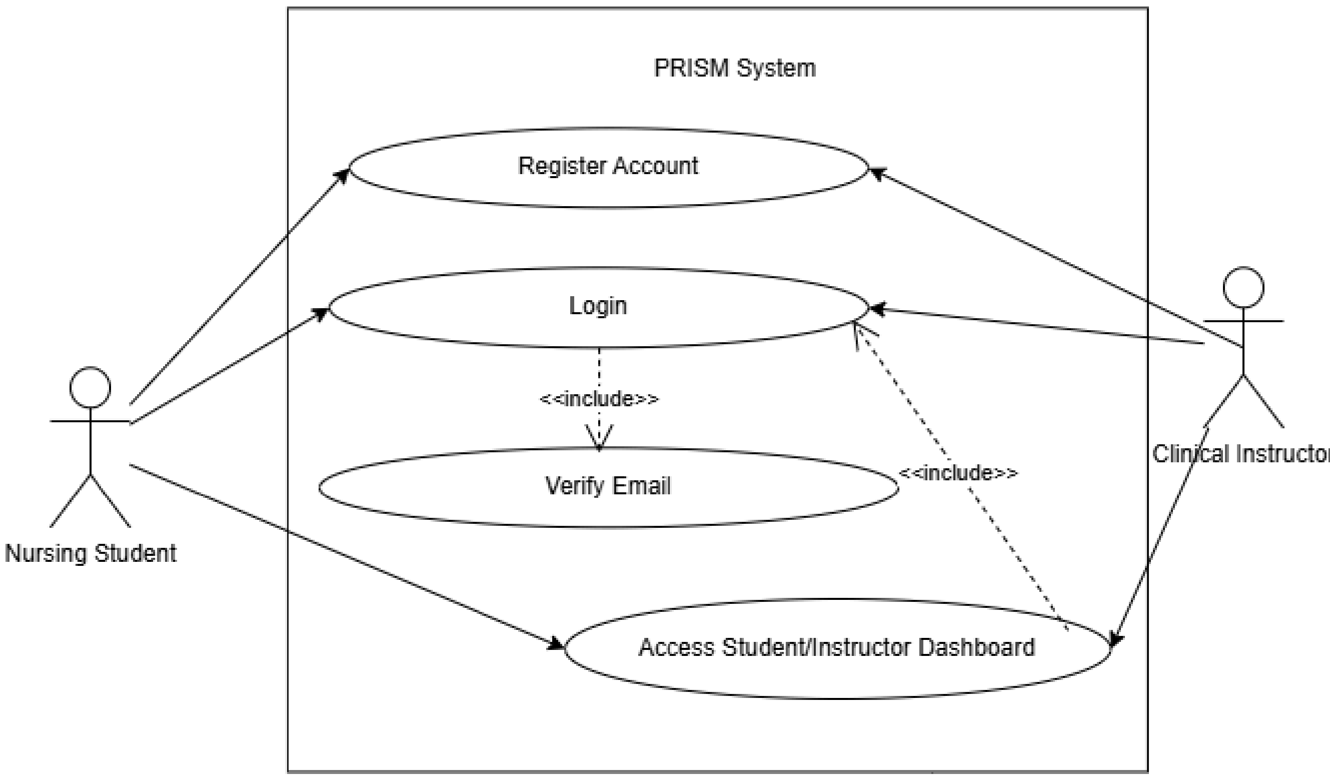
Module 3

Corresponds to General Objective 3: To design and implement a role-based module within PRISM enabling nursing students to access AI-generated feedback with targeted improvement guidance, with all session records stored in Firebase Firestore.

Specific Objectives covered: 1.2 (injection type user input), 3.1 (Firebase Authentication with role-based access control), 3.2 (Firebase Firestore-backed data module for students and instructors), 3.3 (UAT support with ≥5 students and ≥2 instructors).

UC-3.1: User Registration, Login, and Role-based Access

Use Case Diagram

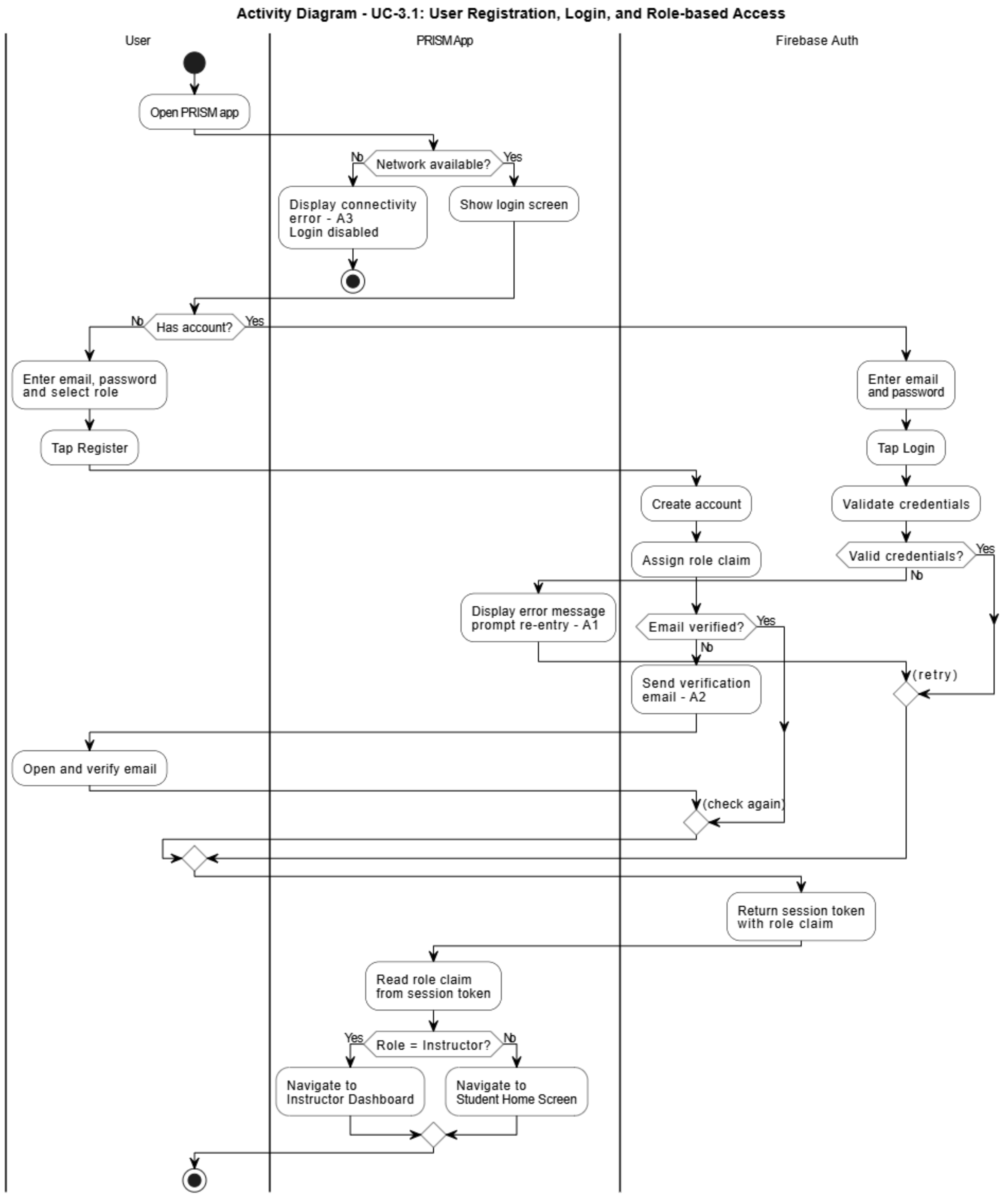


Use Case Description

Use Case ID	UC-3.1
Use Case Name	User Registration, Login, and Role-based Access
Module	Module 3: Role-based Access and Student View
Actors	Nursing Student, Clinical Instructor

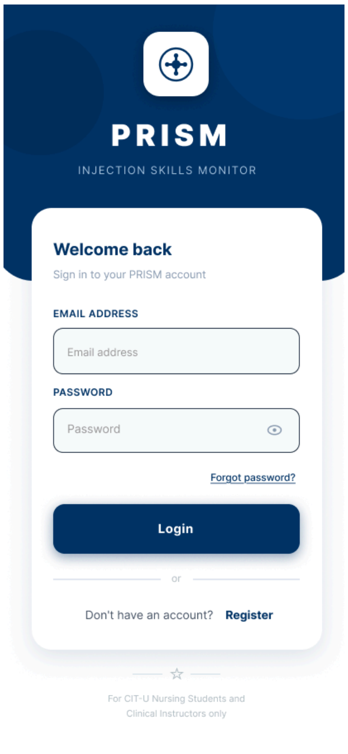
Preconditions	1. PRISM is installed on an Android device. 2. Device has internet connectivity.
Main Flow	1. User opens PRISM and is presented with a login screen. 2. New users select ‘Register’, enter email, password, and role (Student or Instructor). 3. Firebase Authentication creates the account and assigns the role claim. 4. Existing users enter credentials and tap ‘Login’. 5. Firebase Authentication validates credentials and returns a session token. 6. The system reads the role claim and directs the user to the role-specific home screen (Student: Home Screen with ‘Start Session’; Instructor: Dashboard).
Alternative Flow	A1 — Invalid credentials: System displays an error and prompts re-entry. A2 — Unverified email: System prompts email verification before granting access. A3 — Network unavailable: System displays a connectivity error and disables login until connectivity is restored.
Postconditions	User is authenticated with a valid session token and directed to their role-specific interface.
Related SOs	Specific Objective 3.1
Priority	High

Activity Diagram

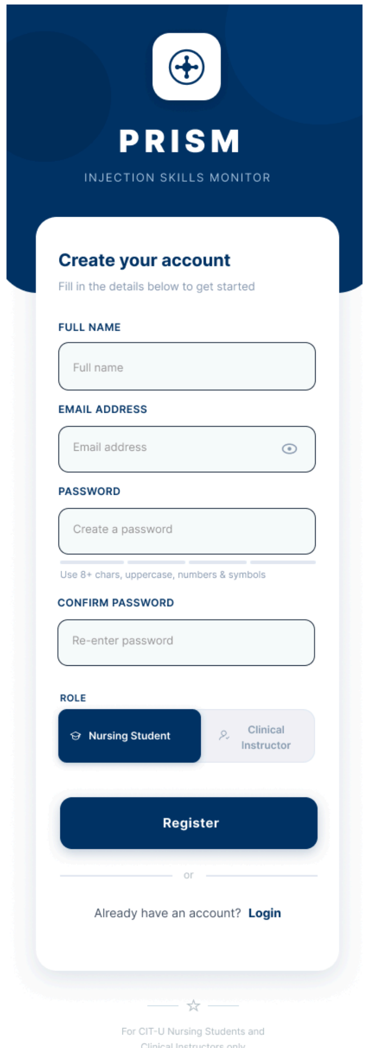


Wireframe

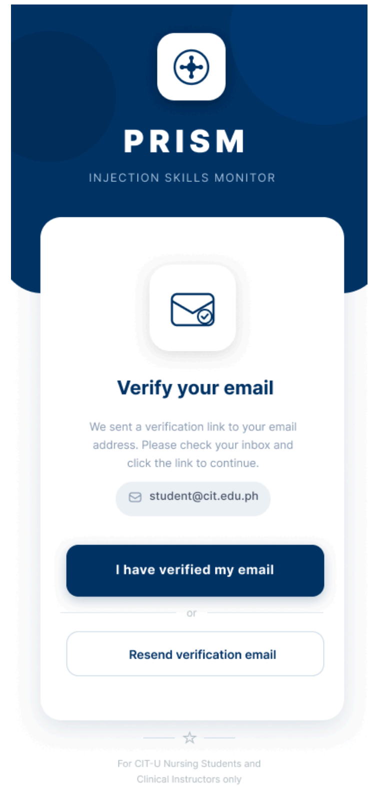
Login Screen



Register Screen

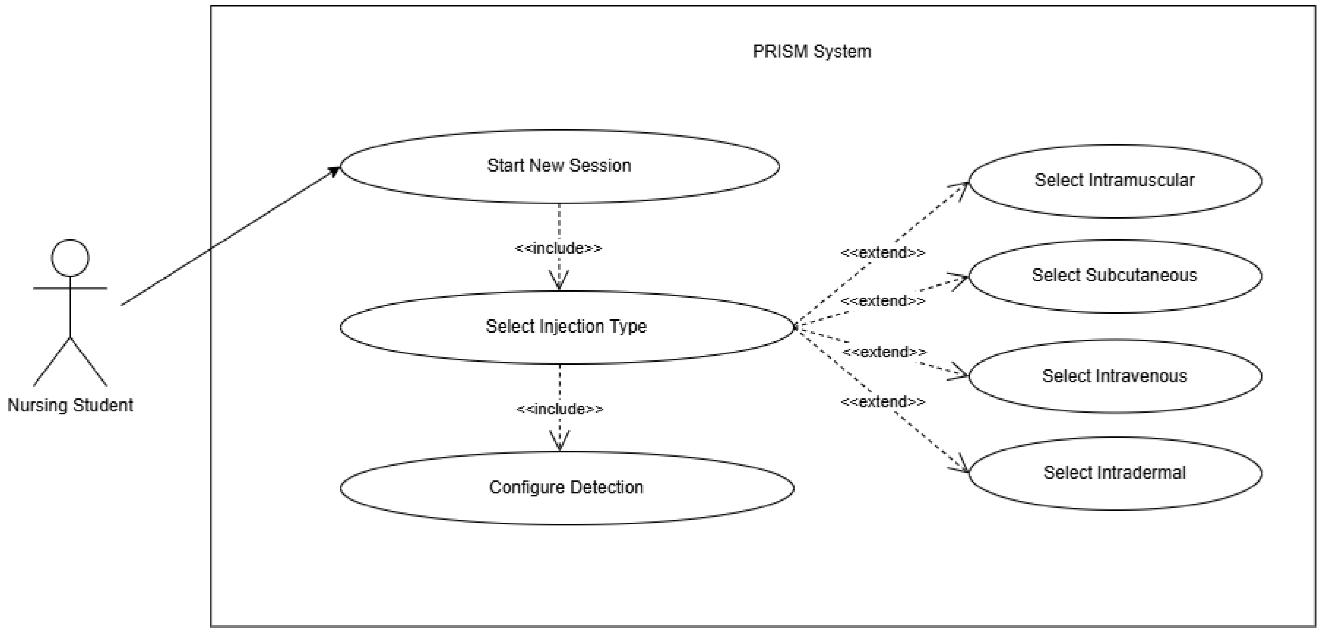


Email Verification Screen



UC-3.2: Injection Type Selection

Use Case Diagram

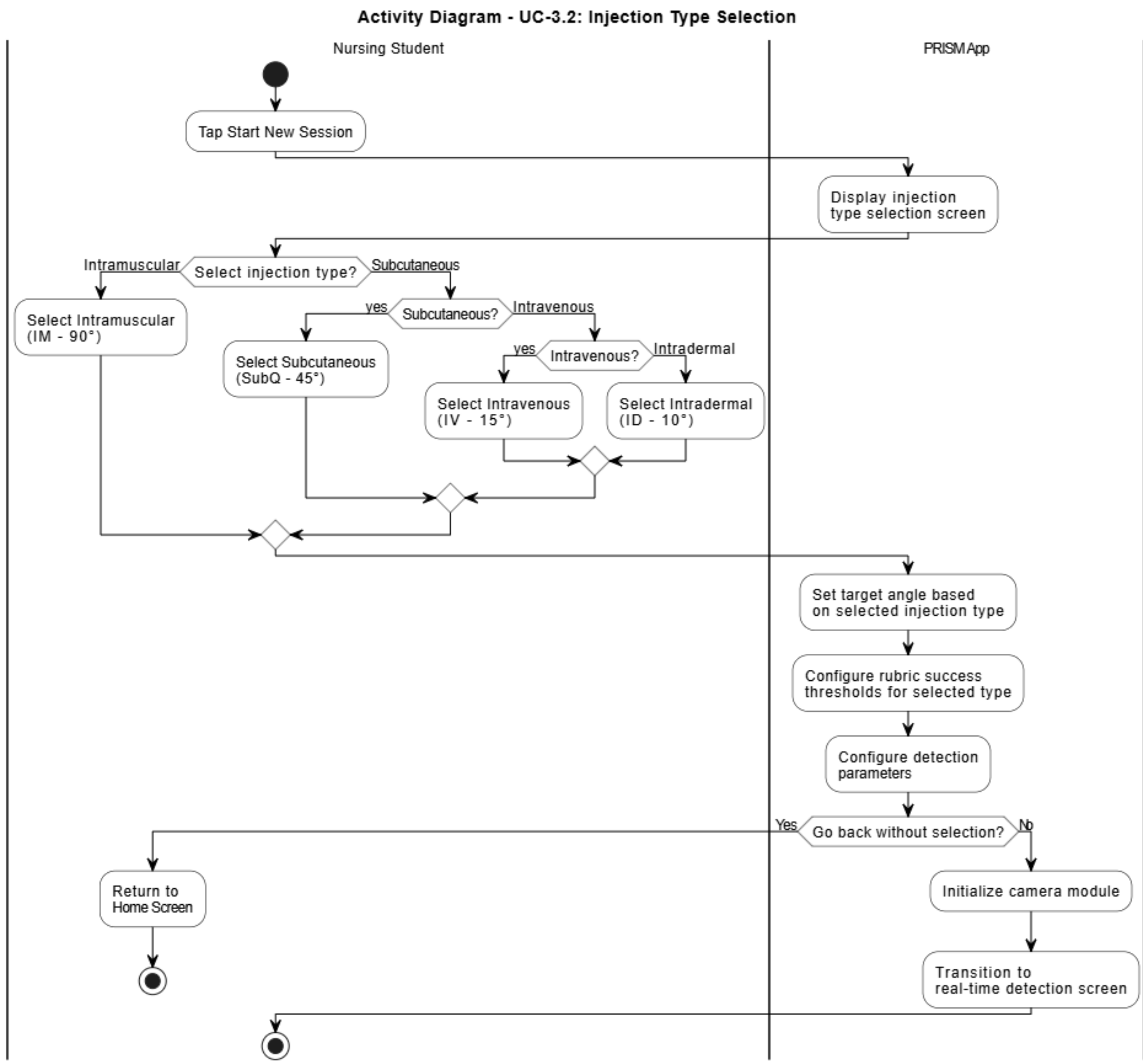


Use Case Description

Use Case ID	UC-3.2
Use Case Name	Injection Type Selection
Module	Module 3: Role-based Access and Student View
Actors	Nursing Student
Preconditions	1. Student is authenticated (UC-3.1) and on the Student Home Screen.
Main Flow	1. Student taps ‘Start New Session’. 2. The system presents four injection type options: Intramuscular (IM, 90°), Subcutaneous (SubQ, 45°), Intravenous (IV, 15°), Intradermal (ID, 10°). 3. Student selects the appropriate injection type. 4. The system configures detection parameters: target angle, rubric success thresholds, and scoring weights for the selected type. 5. The system transitions to the real-time detection screen and activates the camera module (UC-1.1).
Alternative Flow	A1 — Student navigates back without selection: No session is initiated; student returns to Home Screen.
Postconditions	Detection parameters are configured. Camera module is initialized. UC-1.1 begins.

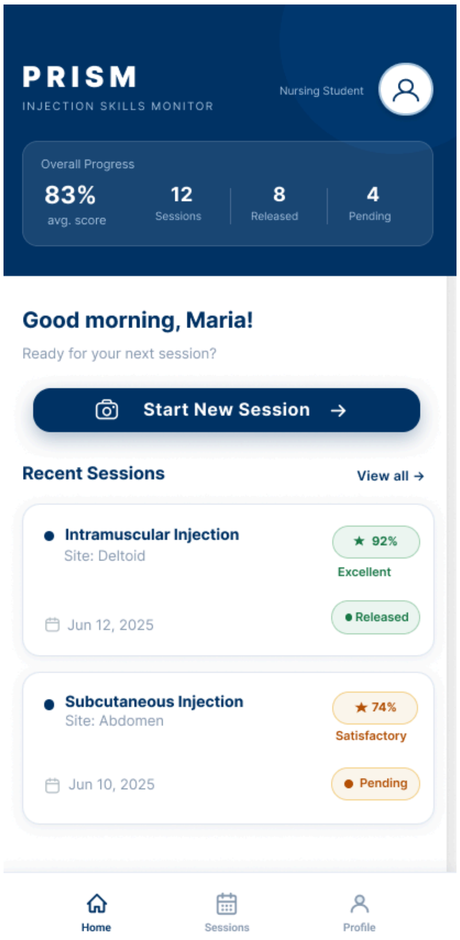
Related SOs	Specific Objective 1.2
Priority	High

Activity Diagram

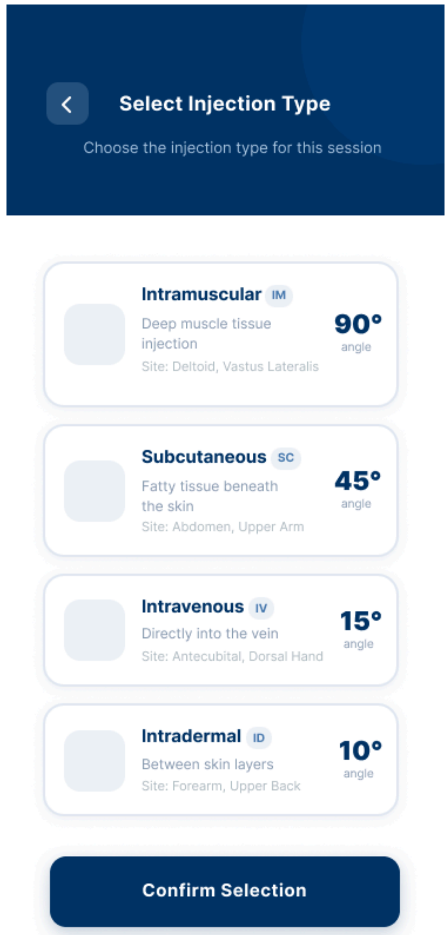


Wireframe

Student Home Screen

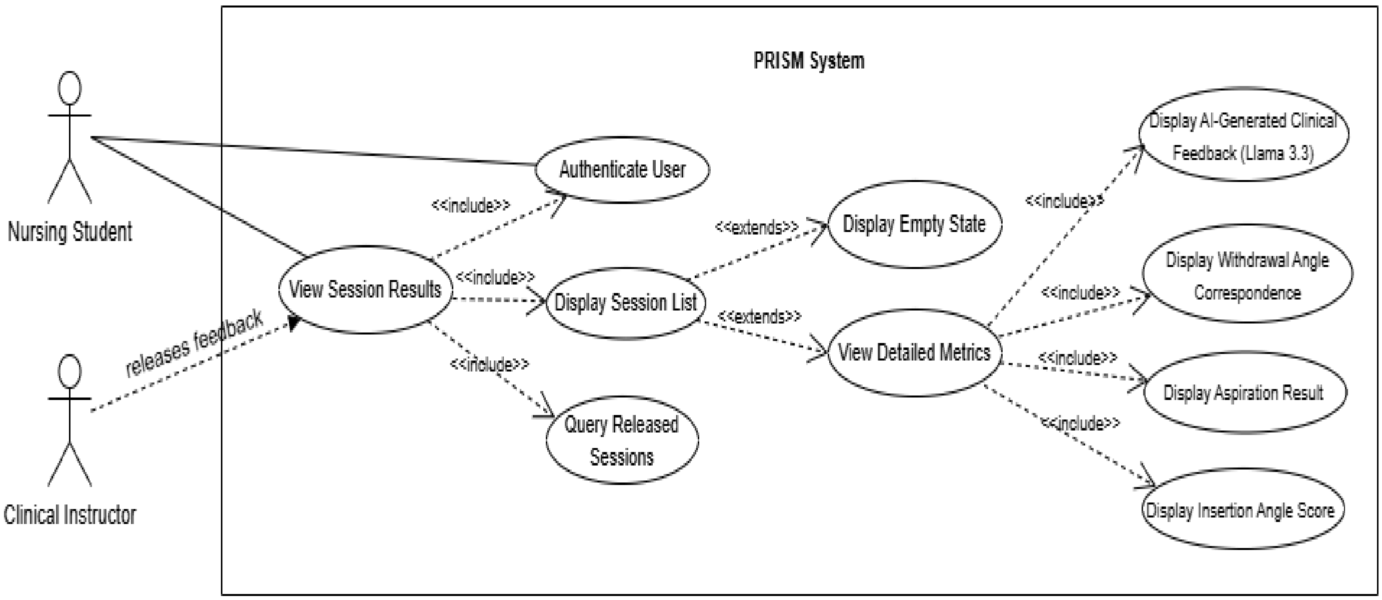


Injection Type Selection Screen



UC-3.3: Student Session Results View

Use Case Diagram

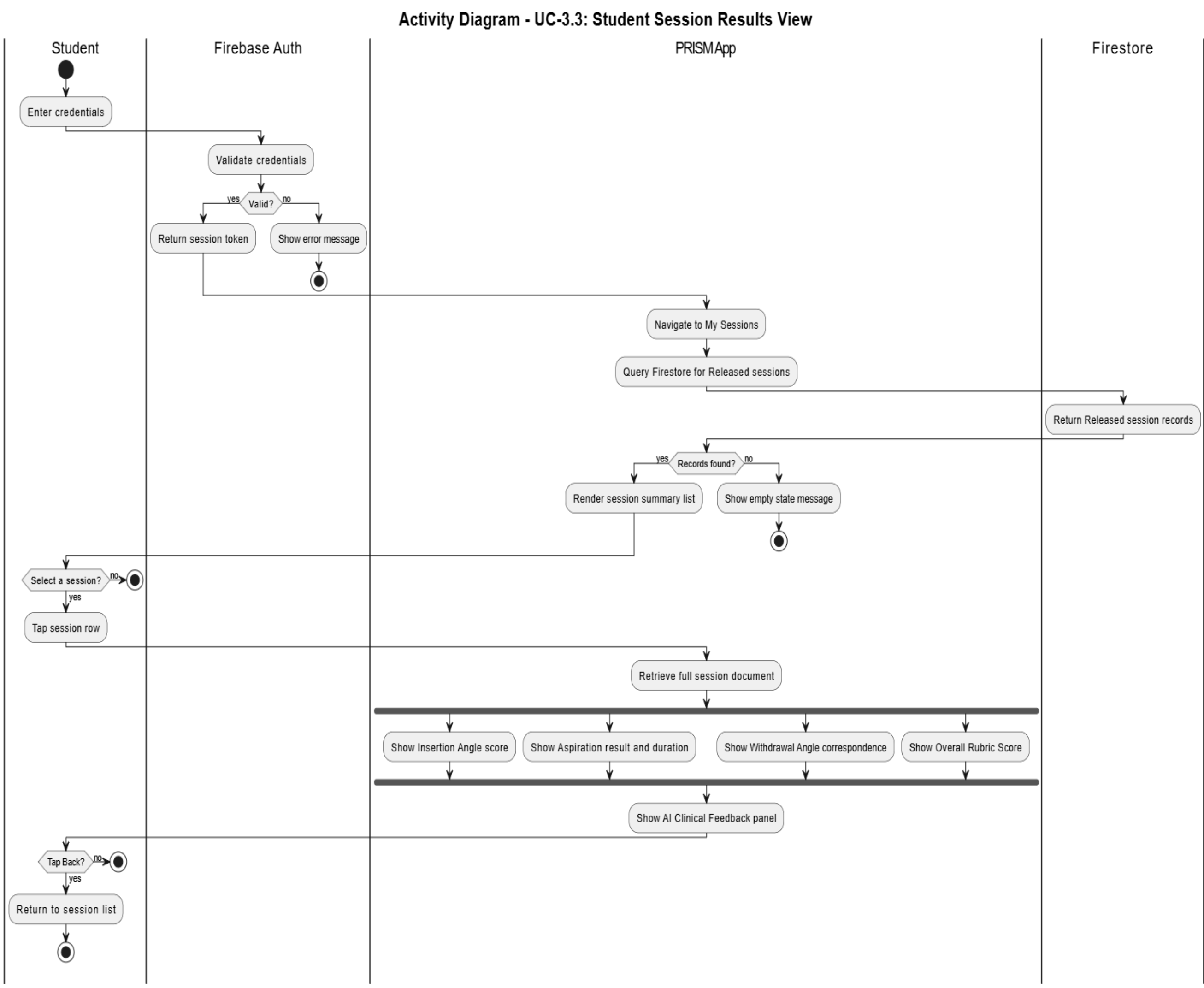


Use Case Description

Use Case ID	UC-3.3
Use Case Name	Student Session Results View
Module	Module 3: Role-based Access and Student View
Actors	Nursing Student
Preconditions	1. Student is authenticated. 2. At least one session has been completed and feedback has been released by an instructor (UC-4.2).
Main Flow	1. Student navigates to ‘My Sessions’. 2. System retrieves all session records from Firestore associated with the authenticated student where feedback status = Released. 3. Student selects a session to view detailed results. 4. The results view displays: injection type, insertion angle value and rubric score, aspiration detection result and metrics, withdrawal angle value and correspondence score, overall rubric score, and AI-generated clinical feedback with targeted improvement guidance. 5. Student can scroll through historical sessions sorted by date (most recent first).
Alternative Flow	A1 — No released sessions: System displays ‘No results available yet. Your instructor will release feedback after review.’

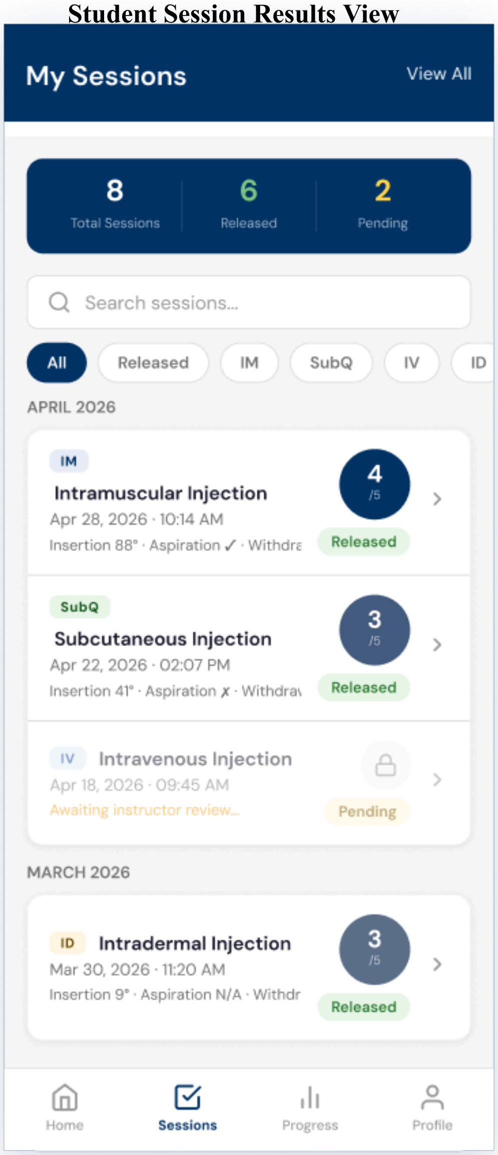
Postconditions	Student has reviewed session results and AI-generated improvement guidance.
Related SOs	Specific Objective 3.2
Priority	High

Activity Diagram

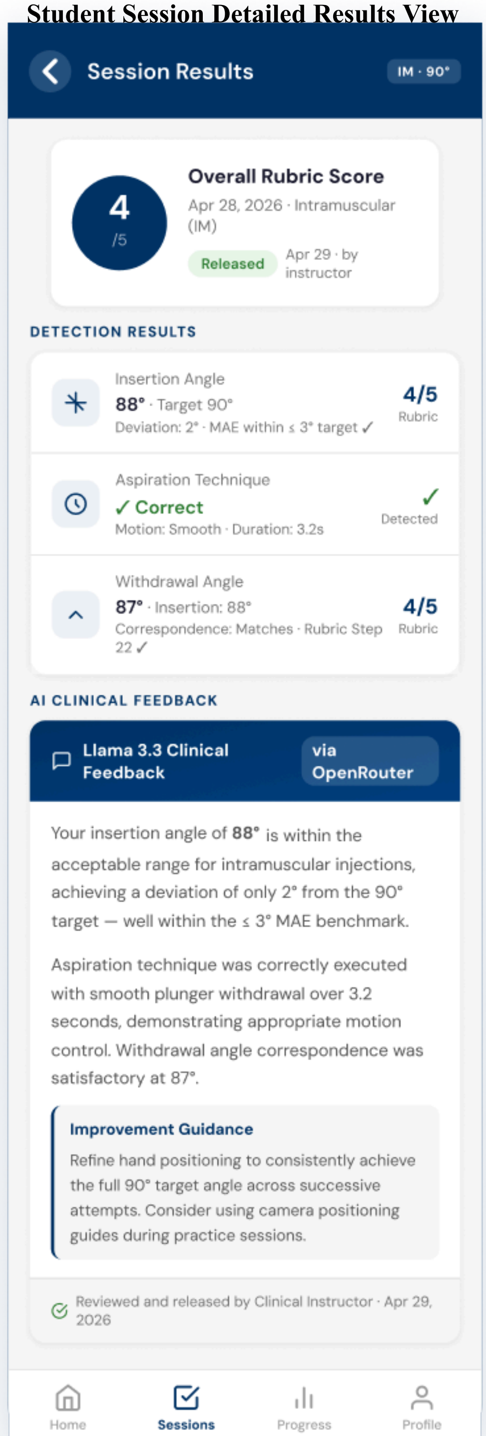


Wireframe

Student Session Results View



Student Session Detailed Results View



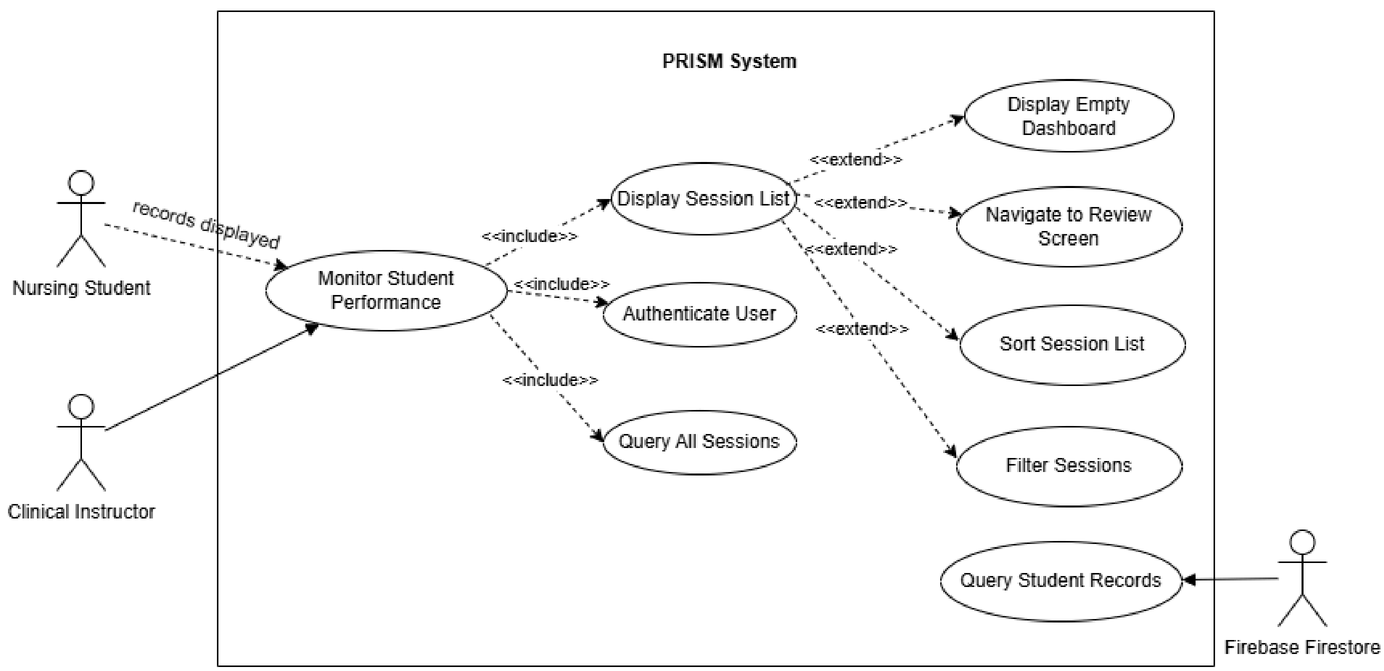
Module 4

Corresponds to General Objective 4: To develop a feedback review module within PRISM that allows clinical instructors to review and refine AI-generated feedback before it is released to nursing students.

Specific Objective covered: 4.1 (instructor feedback review and release via Firebase Firestore-backed interface).

UC-4.1: Instructor Performance Monitoring Dashboard

Use Case Diagram

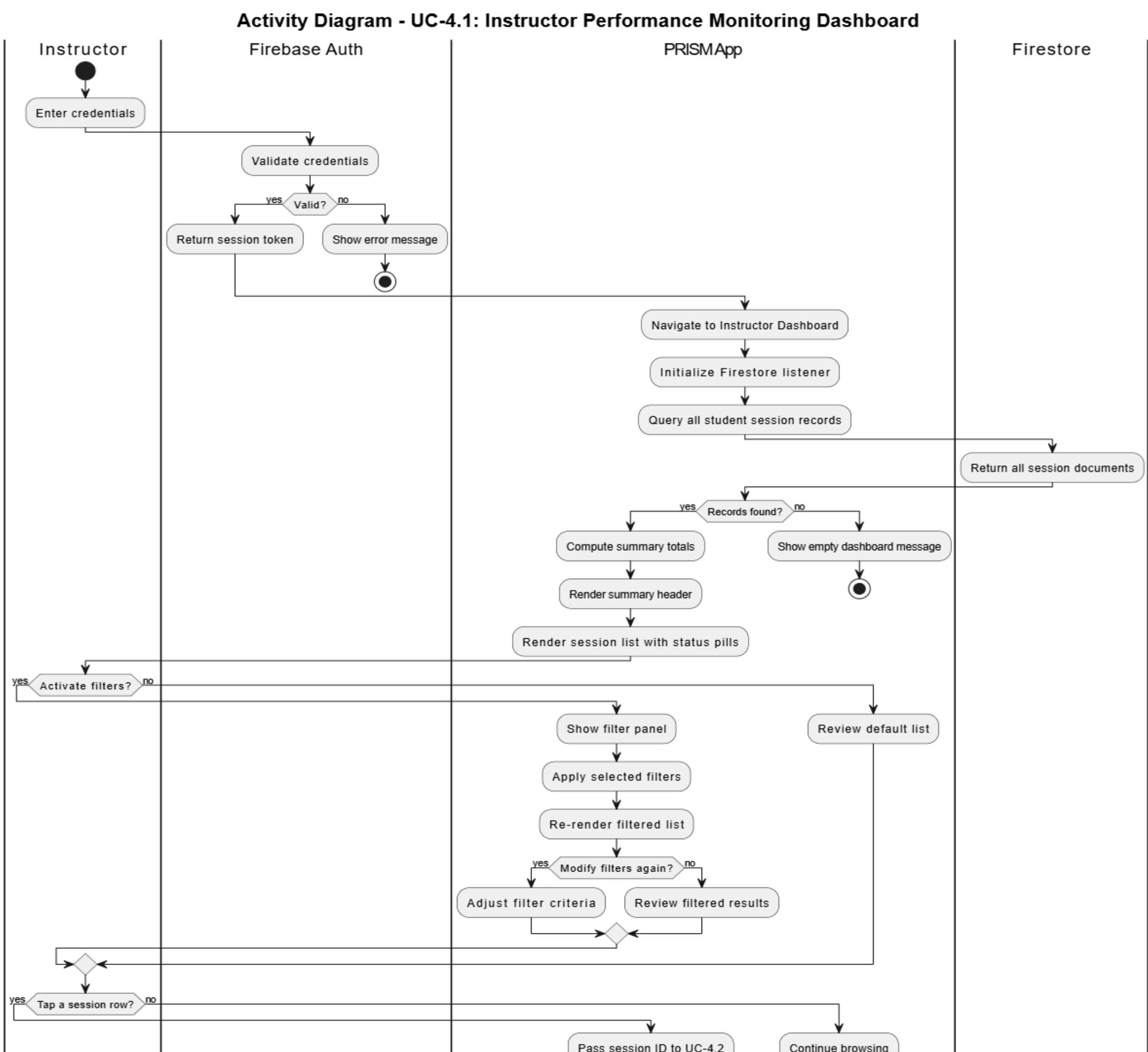


Use Case Description

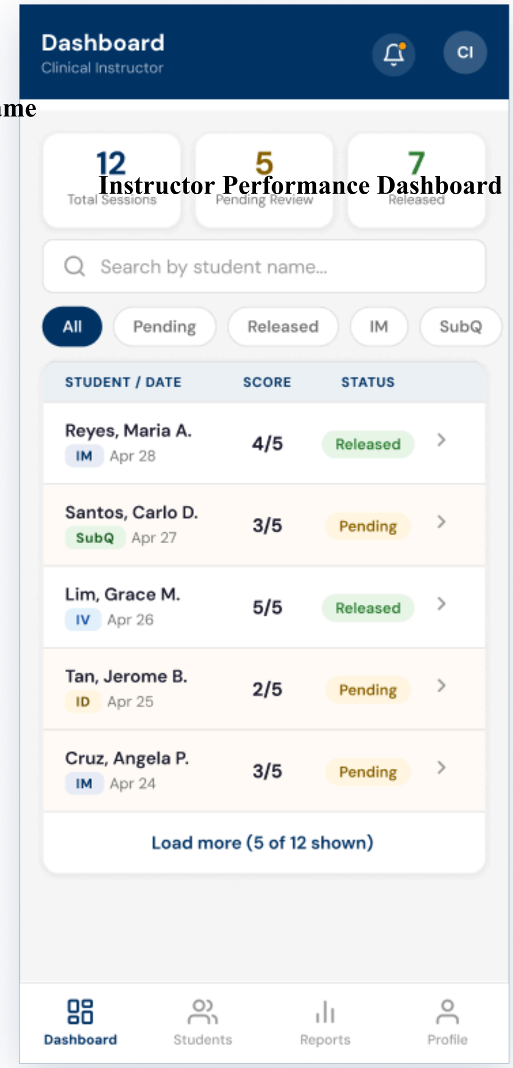
Use Case ID	UC-4.1
Use Case Name	Instructor Performance Monitoring Dashboard
Module	Module 4: Instructor Dashboard and Feedback Review
Actors	Clinical Instructor
Preconditions	1. Instructor is authenticated (UC-3.1) with an Instructor role claim.

Main Flow	1. Instructor is directed to the Dashboard upon login. 2. System retrieves all student session records from Firestore accessible to the authenticated instructor. 3. Dashboard displays a list view of students showing: student name, most recent session date, injection type, overall rubric score, and feedback status (Pending / Released). 4. Instructor can filter the list by student name, injection type, date range, or status. 5. Instructor taps a session entry to navigate to the Feedback Review screen (UC-4.2).
Alternative Flow	A1 — No sessions recorded: Dashboard displays ‘No student sessions found.’
Postconditions	Instructor has an overview of all student session records and can navigate to individual feedback review.
Related SOs	Specific Objective 4.1
Priority	High

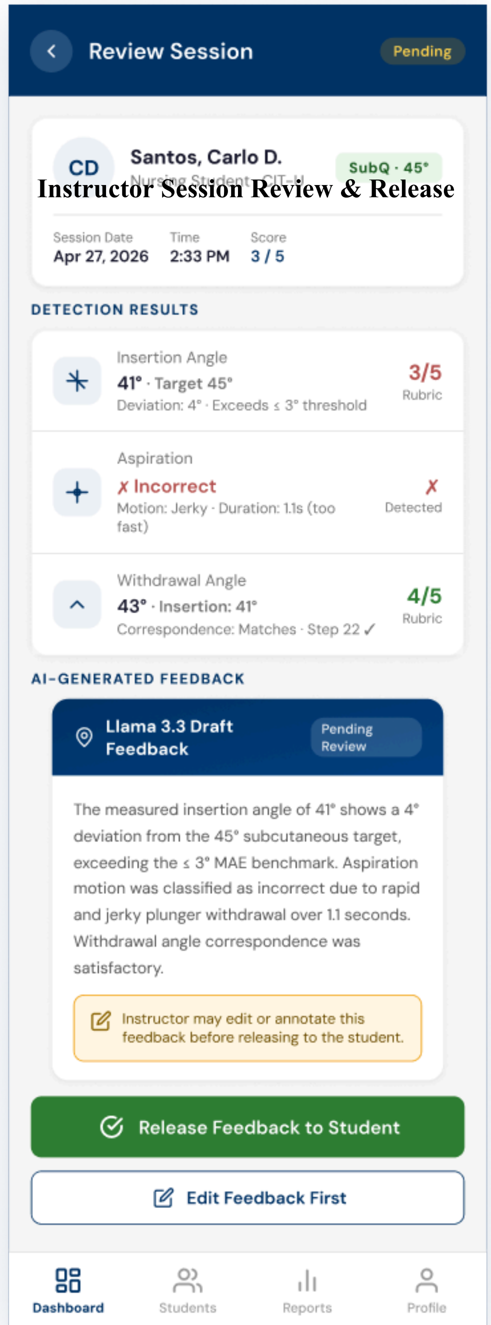
Activity Diagram



Wireframe



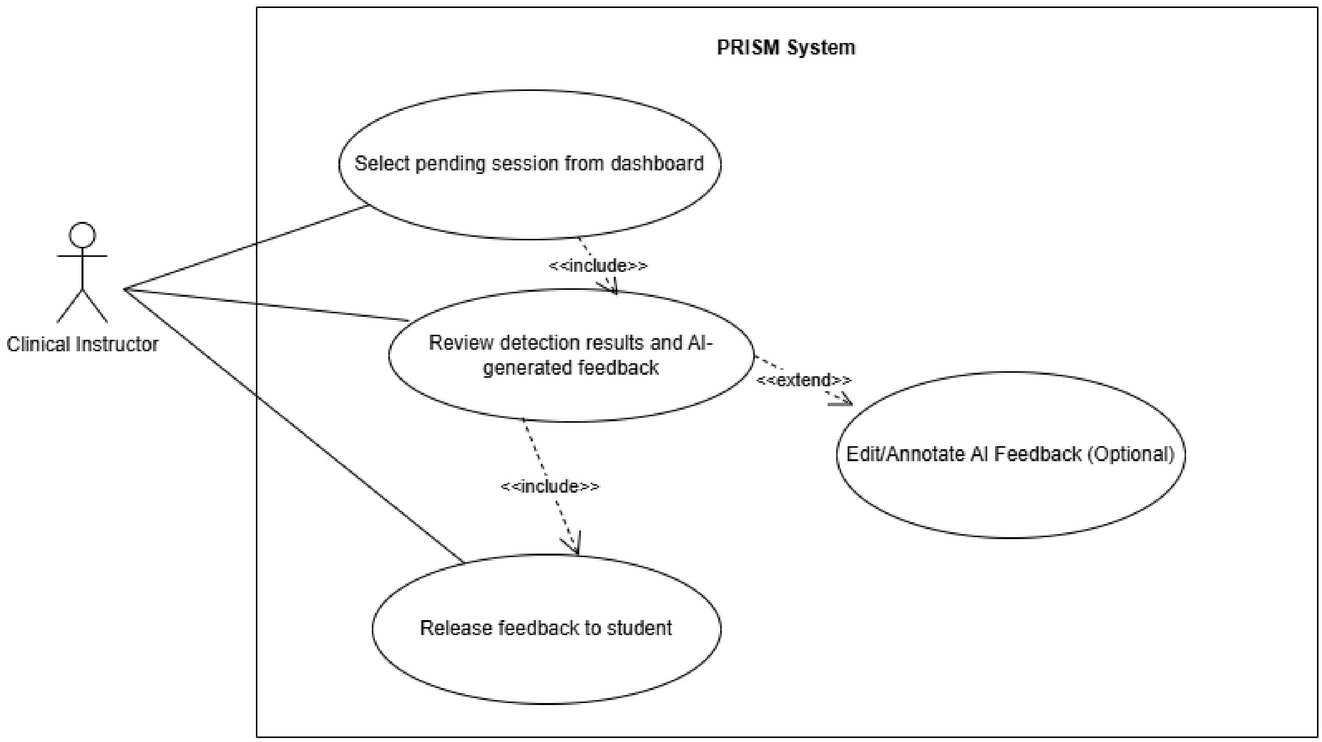
Instructor Performance Dashboard



Instructor Session Review & Release

UC-4.2: AI Feedback Review and Release

Use Case Diagram



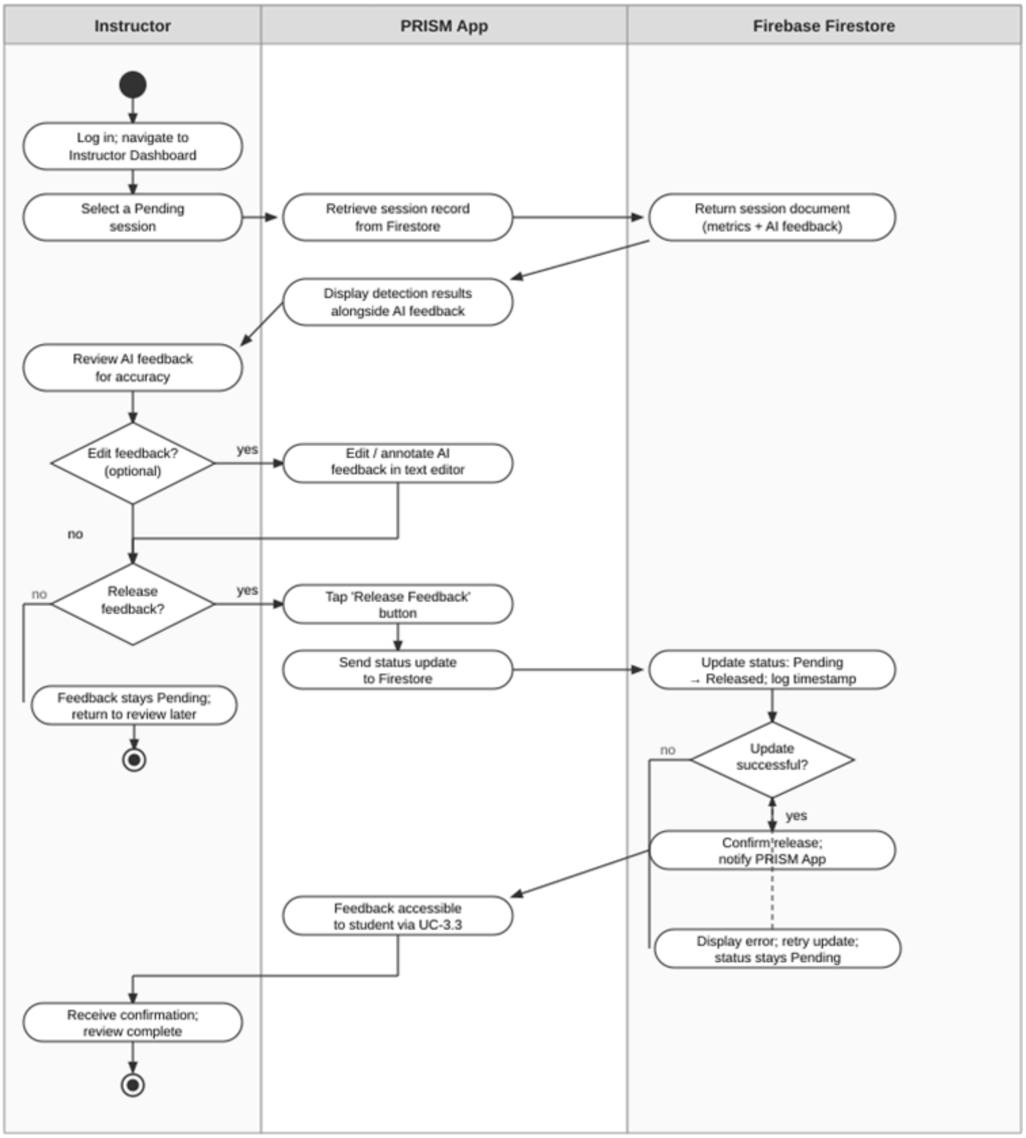
Use Case Description

Use Case ID	UC-4.2
Use Case Name	AI Feedback Review and Release
Module	Module 4: Instructor Dashboard and Feedback Review
Actors	Clinical Instructor
Preconditions	1. Instructor is authenticated. 2. At least one session with Pending feedback status exists in Firestore.

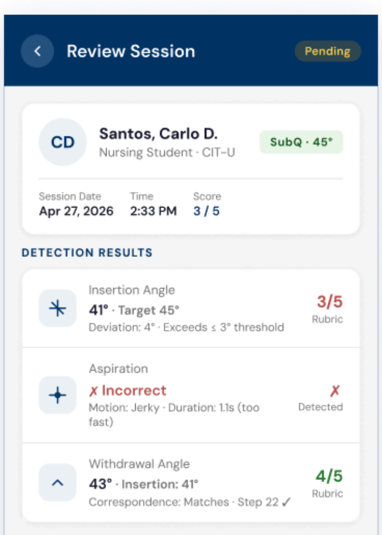
Main Flow	1. Instructor selects a Pending session from the Dashboard (UC-4.1). 2. System displays the full detection results alongside the AI-generated feedback text. 3. Instructor reviews the feedback for clinical accuracy and appropriateness. 4. Instructor may optionally edit or annotate the AI-generated feedback text using an in-app text editor. 5. Instructor taps ‘Release Feedback’. 6. System updates the Firestore session document: feedback status changes from Pending to Released; release timestamp is recorded. **If the instructor modified the text box during Step 4, the system triggers an auditing subroutine: setting an is_edited_by_instructorflag totrueand caching the pristine, unedited Llama 3.3 string inside anoriginal_ai_feedback document attribute. 7. Feedback becomes accessible to the student in UC-3.3.
Alternative Flow	A1 — Instructor chooses not to release: Feedback remains Pending; instructor may return to review later. A2 — Firestore update fails: System displays an error and retries. **All modification flags and text states are held securely in application memory;** feedback status remains Pending until write confirmation is verified by the cloud database layer.
Postconditions	Session feedback status is Released in Firestore. Student can access the validated feedback via UC-3.3.
Related SOs	Specific Objective 4.1
Priority	High

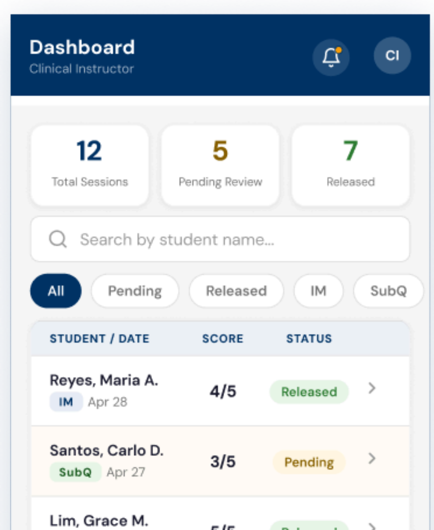
Activity Diagram

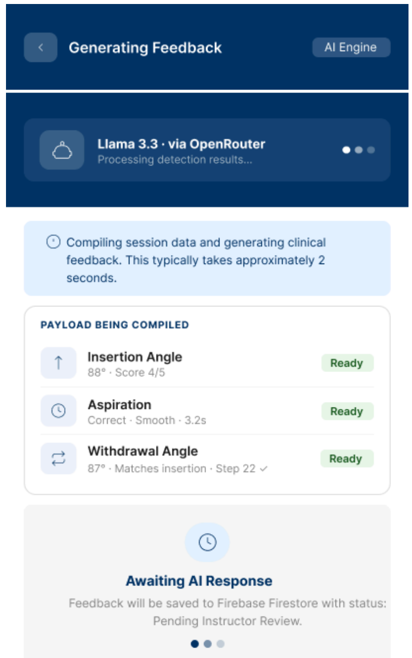
Activity Diagram - UC-4.2: AI Feedback Review and Release



Wireframe







3.4 Non-functional requirements

Performance

- MediaPipe Hand Landmarker must process each camera frame and return landmark coordinates within 100 ms to support perceptually real-time angle overlay display.
- Needle insertion angle detection must achieve a Mean Absolute Error (MAE) of ≤ 3° relative to expert-annotated baselines, consistent with the clinical threshold established by Walsh et al. (2021).
- AI feedback generation via the OpenRouter API (Llama 3.3 70B Instruct) must return a complete, structured response within approximately 2 seconds of the API request being sent.
- Firebase Firestore read and write operations (session record storage and retrieval) must complete within 3 seconds under a stable 4G LTE or Wi-Fi connection.
- The application must reach a usable state (login screen displayed) within 5 seconds of launch on a device meeting the minimum hardware specifications (ARM64 CPU, 3 GB RAM, Android 8.0+).
- The system must achieve a System Usability Scale (SUS) score of ≥ 80 ("Good" classification) during UAT with a minimum of 5 nursing students and 2 clinical instructors.

Security

- All user authentication must be handled exclusively by Firebase Authentication. Role claims (Student or Instructor) are assigned at registration and validated server-side on every authenticated request.
- Students may only access their own session records. Clinical instructors may access all session records within their assigned scope. No user may self-escalate their role.
- All communications between PRISM and external services (Firebase Authentication, Firestore, OpenRouter API) must use HTTPS with TLS 1.2 or higher. No plaintext transmission of credentials or session data is permitted.
- API keys for OpenRouter and Firebase must be stored in environment configuration and must never be hardcoded in application source code or exposed in client-side builds.
- No session video footage is stored at any point. Only computed metrics (angles, aspiration result, rubric scores) and AI-generated feedback text are persisted in Firestore.
- Firestore security rules must enforce that session documents with Pending feedback status are not readable by students. Feedback becomes readable to the student only after an authenticated instructor sets the status to Released (UC-4.2).
- On logout, the Firebase Authentication session token must be invalidated and local authentication state cleared. No credentials are stored in plaintext in local storage or shared preferences.

Reliability

- If MediaPipe loses hand landmark tracking during a session (due to occlusion or poor lighting), the system must display a ‘Detection Lost’ warning immediately, pause angle recording, and automatically resume recording upon landmark re-acquisition without requiring a session restart (UC-1.1, Alternative Flow A1).
- If the OpenRouter API times out or returns a malformed response, the system must retry once. On persistent failure, raw detection metrics must be saved to Firestore and the session flagged with a ‘Feedback Generation Failed’ status for manual instructor follow-up. No detection data must be lost as a result of an API failure (UC-2.1, Alternative Flows A1 and A2).
- All Firestore session record writes must be atomic. Partial writes resulting in incomplete session documents are not acceptable. If a Firestore write fails during feedback release (UC-4.2), the system must retry and display an error to the instructor; the status must not be set to Released unless the update is confirmed.
- The on-device MediaPipe detection pipeline must remain functional without internet connectivity, allowing students to complete the procedural detection phase during temporary network loss. Cloud-dependent features (Authentication, Firestore, OpenRouter) must be disabled gracefully with a clear connectivity error message rather than crashing.
- All system errors (API failures, Firestore write failures, authentication errors, detection loss) must surface to the user through clear, non-technical, actionable in-app messages. No error condition defined in the use case alternative flows must result in data loss or an unrecoverable application state.
- Session state (insertion angle, aspiration result, withdrawal angle) must be maintained in memory for the full duration of a session and must not be lost due to brief Android background transitions. All session state must be successfully committed to Firestore before AI feedback generation is triggered.